



Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: A vortex-dynamics approach

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ABSTRACT

Two vortex-induced vibration (VIV) events of the main girder of long-span bridges successively happened on the Yingwuzhou suspension bridge and Humen suspension bridge in China, 2020, and drew public attention. Especially for the Humen suspension bridge, the amplitude of VIV was so notable that the traffic was closed for 10 days. Generally, VIV is sensitive to the structural and ambient issues and large-amplitude VIV is unacceptable for engineering structures as it could result in potential fatigue failure of some key joint components. Therefore, it is of great significance to explore the excitation mechanism of VIV and develop effective methods to mitigate excessive vibration of main-girders. From the perspective of vortex dynamics, we present a selective review of recent progress on the mechanism of VIV occurred in long-span bridges and aerodynamic countermeasures to attenuate the VIV amplitude. Notwithstanding the VIV mechanism of girders has not been thoroughly clarified, it is generally accepted that VIV rises from the unsteady vortex shedding from the main girder. Besides, the flow control methods, which are proposed to manipulate the surrounding flow patterns around the girder, are also focused in this review. Some perspectives are finally presented for future studies in fluid–structure interaction and aerodynamic vibration mitigation of long-span bridges.

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1. Introduction

Long-span bridges are prone to ambient excitations, e.g., wind, rain, and vehicles, due to their notable flexibilities and low damping ratios. When wind passes the main-girders of long-span bridges, flow separation occurs at the leading and trailing edges, and the boundary layers are rolled up to form vortices shedding alternatively into the wake. If the shedding frequency of the wake vortex approaches some-order modal frequency of the girder, a flow–structure coupling oscillation, namely, vortex-induced vibration (hereinafter VIV), builds up with a relatively high amplitude. One of the most notable characteristics of VIV is that the wake shedding frequency would approach the frequency of the girder in a limited range of wind velocity. This process is often referred to as *lock-in* (Blevins and Vibrations, 1990; Fredsoe

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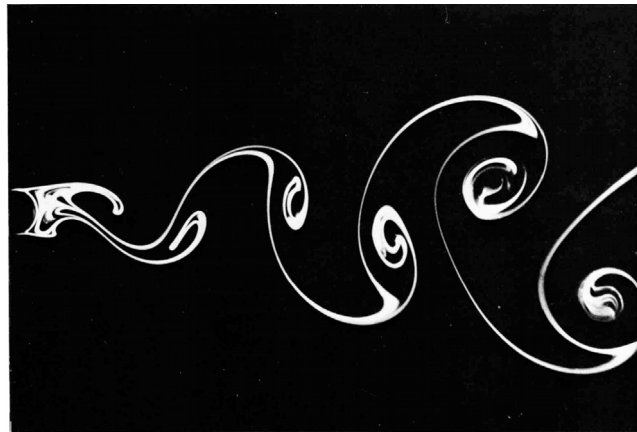


Fig. 1. The vortex shedding phenomenon in the cylinder wake at $Re = 140$ (Van Dyke, 1982).

and Sumer, 1997). Some representative VIV cases have been reported on a number of bridges, for example, the Storebælt suspension bridge (Larsen, 1993), Rio–Niterói bridge (Battista and Pfeil, 2000), Trans-Tokyo Bay (Fujino and Yoshida, 2002), Alconétar Arch bridge (Astiz, 2010), Xihoimen suspension bridge (Li et al., 2011a) and Volgograd Bridge (Corriols, 2016). The excessive VIV is unacceptable in engineering field under consideration of possible physiological impacts on drivers crossing the bridge and potential fatigue failure of some key joint components of bridges from a structural safety point of view.

To understand the excitation mechanism of VIV of bridge girders, the VIV of a circular cylinder is discussed first. Though a circular cylinder has a quite simple geometry shape, flow around a circular cylinder is called kaleidoscope in the field of fluid mechanics, for it contains rich and complex flow phenomena (Ferrari et al., 2020). Thus, flow around a circular cylinder has been a research hotspot to engineers as well as to scientists for more than a century. When a circular cylinder is subject to flow, a complex interaction of three shear layers, i.e., boundary layer, separating free shear layers and the wake, leads to the alternating *von Kármán vortex street*, as shown in Fig. 1. These shedding vortices apply dynamic loads on the cylinder surface, and may cause transverse VIV when the vortex shedding frequency matches the structural frequency, as mentioned above. To suppress the vortex shedding behind a circular cylinder and thus to mitigate VIV, a great number of active (Yang et al., 2021; Chen et al., 2014a; Feng et al., 2010; Chen et al., 2013; Gao et al., 2019a) and passive control schemes (Zhou et al., 2019; Xia et al., 2018; Owen et al., 2001; Bearman and Branković, 2004; Law and Jaiman, 2018; Gao et al., 2017b,a; Chen et al., 2015; Baek and Karniadakis, 2009; Gao et al., 2019b) have been proposed. Previous studies in vortex dynamics around a circular cylinder and flow control methods have been of valuable references for bridge aerodynamics and aerodynamic countermeasures.

Lock-in is a distinct feature of VIV (Wu et al., 2012). With the augment of incoming flow speed, the vortex shedding frequency (f_s) of a fixed cylinder follows the Strouhal law ($St = f_s D/U$, in which D is the cylinder diameter and U is the flow speed). While for the cylinder elastically supported in transverse direction, the *lock-in* phenomena occur when f_s matches the natural frequency of the cylinder, where large-amplitude VIV is developed. de Langre and colleagues (de Langre, 2006; Violette et al., 2010; Facchinetti et al., 2004a,b; Mathelin and De Langre, 2005) developed a framework of linear stability analysis to reveal the essence of the lock-in during VIV. The instability of lock-in is referred to as coupled-wake flutter, i.e., the wake mode is coupled with the structure mode within the lock-in region. In the initial evolution stage, the wake mode and the structure mode co-exist accompanied by a competition phenomenon. As time goes on, the dynamic coupling system is gradually governed by the structure mode, resulting in the frequency synchronization of structural vibration and vortex shedding. Though the linear stability has been validated by experiments at relatively high Reynolds number (Re) (For instance, Gao (2019)), a recent numerical simulation by Zhang et al. (2015) presented a more-detailed investigation of the lock-in scenario during VIV, where the frequency lock-in at low Re can be categorized into resonance- and flutter-induced ones based on different potential mechanisms. As the natural frequency of the structure matches the eigenfrequency of the uncoupled wake mode, the coupled system is governed by resonance instability. Whereas the structure frequency becomes slightly larger, coupled-mode flutter between structure mode and wake mode turns to be dominant.

The long-span bridges are built to provide transportation, thus the plate-like cross-section is usually the first choice of the main girders due to its economical and convenient implementation. Fig. 2 presents the smoke-wire visualization of the flow past a rectangular prism with a aspect ratio of 10. The unsteady shear layers separate at the leading edges and reattach on the plate surface. As the reattaching flow moves downstream, alternating vortex shedding develops behind the structures. For a rectangular plate, more vortex mechanisms of VIV could be obtained according to the different aspect ratio and Re . As the aspect ratio is increased, three types of vortex formation, i.e., leading-edge vortex shedding (LEV), impinging leading-edge vortices (ILEV) and trailing-edge vortex shedding (TEV), are observed respectively. With

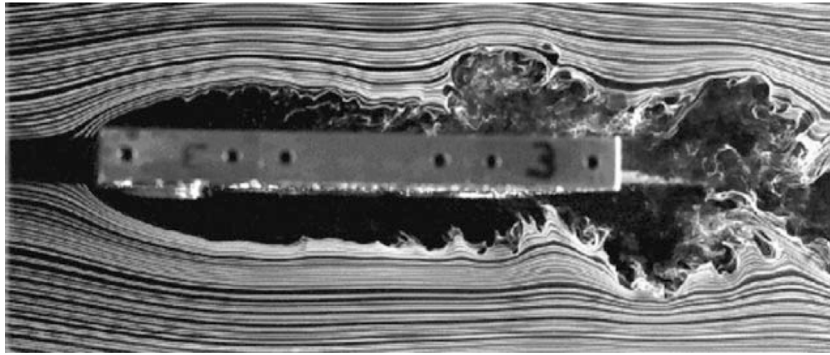


Fig. 2. Dynamic separation and vortex shedding from a rectangular prism with a chord ratio of 10 ($Re = 8667$, flow is visualized by smoke-wire method) (Mills et al., 2002).

the increasing Re , the boundary value of aspect ratio between then is slightly postponed due to the increasing wind velocity. As Re grows to a certain value and continues to increase, the Strouhal number (St) will remain constant. With regard to ILEV, the increasing Re will shift the transition point upstream, reduce the sensitivity of shear layer to external disturbances and increase the three-dimensional instability of the flow along the plate sides (Mills et al., 2002). Besides, the vortex formation is sensitive to external perturbations (Parker and Welsh, 1983), which may rise from the buffeting in its natural frequency. The motion-induced vortices (MIVs), originating from this excitation, interact with the gradually developed flow structure due to the structure motion. As a result of this dynamic interaction, both the pressure field and aerodynamic forces on the plate are changed. This brings more difficulties to investigate the flow-induced motions of plate-like structures.

To further understand the fluid–structure–interaction phenomenon and the excitation mechanism of VIV of main-girder for long-span bridges, more attention of research is desired, especially from the perspective of vortex dynamics. Besides, as the surrounding flow and vortex pattern are highly correlated to VIV, the vortex dynamics-based insights are also helpful to develop aerodynamic countermeasures to mitigate the VIV amplitudes by manipulating the flow. Inspired by this, the paper is organized as follows: Section 2 reviews three well-known VIV cases of long-span bridges and analyze the in-depth vortex dynamic-based mechanism of vibration. Section 3 and Section 4 elaborate respectively the effectiveness and mechanism of several kinds of passive control and active control methods. Some conclusions are summarized in Section 5, and prospects of VIV control of long-span bridges are also discussed.

2. Typical VIV cases of long-span bridges

In this section, the aspect and excitation mechanism of VIV of the Storebælt and Xihoumen suspension bridge are discussed first based on various approaches including on-site monitoring, wind tunnel experiments and CFD (Computational Fluid Dynamics) simulations.

2.1. Storebælt suspension bridge: Single box girder

The Storebælt suspension bridge (Fig. 3) in Denmark is a remarkable VIV case in modern suspension bridges. The availability of considerable model-based numerical, experimental in-situ full-scaled measurement reports makes it notable for studying.

It is a three-span single-box girder bridge acrossing the Storebælt Strait, Denmark, with a central span length of 1624 m and side spans of 535 m, as sketched in Fig. 3. During the final stages of construction, low-frequency vibrations were observed in 1997/1998. A structural monitoring system consisting of four accelerometers and a data acquisition system was installed in the bridge January 1998 and indicated that the main girder of Storebælt suspension bridge experienced vertical vibrations mainly at its fifth-modal frequency, combined with its third- and six-modal components during the period between 14 Jan 1998 and 16 May 1998 (Larsen et al., 2000). Those vertical VIV was found to happen under moderate wind velocities (5–10 m/s) nearly perpendicular to the bridge axis, and low incoming turbulence intensity (I_u : 0.02–0.06). Furthermore, simultaneous measurements of pressure and acceleration on the Storebælt suspension bridge were subsequently carried out by Frandsen (2001a) who proposed that when VIV occurred, a high correlation between pressure and acceleration within lock-in range was observed than for outside of lock-in.

Frandsen (2000) utilized the finite element method (FEM) and discrete vortex method (DVM) to numerically investigate the Navier–Stokes equations for a fixed and oscillating girder of the Storebælt suspension bridge. The oscillating girder was modeled as a single-degree-of-freedom system with a spring stiffness and mass concentrated at the gravity center. The vertical VIV occurred on the main girder with an amplitude of 0.24 m at $Re = 1.65 \times 10^7$, as illustrated in Fig. 4. The

Main Particulars:

Main Span Length	$l_m = 1624$ m
Side Span Length	$l_s = 535$ m
Cable Sag	$d = 180$ m
Cable Spacing	$2a = 27.0$ m
Cable Area (one cable)	$A_c = 0.44$ m ²
Girder Width	$B = 31.0$ m
Girder Depth	$D = 4.4$ m
Girder Mass (incl. cable)	$m = 22.74 \cdot 10^3$ kg/m
Girder Mass Moment of Inertia (incl. cable)	$I = 2.47 \cdot 10^6$ kgm ² /m

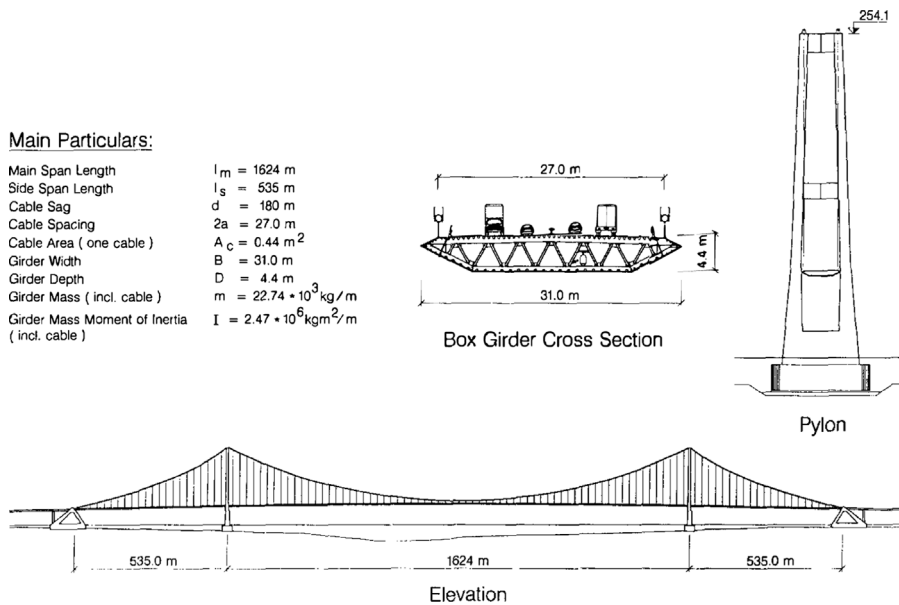


Fig. 3. Detailed arrangement of the Storebælt suspension bridge and its cross-section plan (Larsen, 1993).

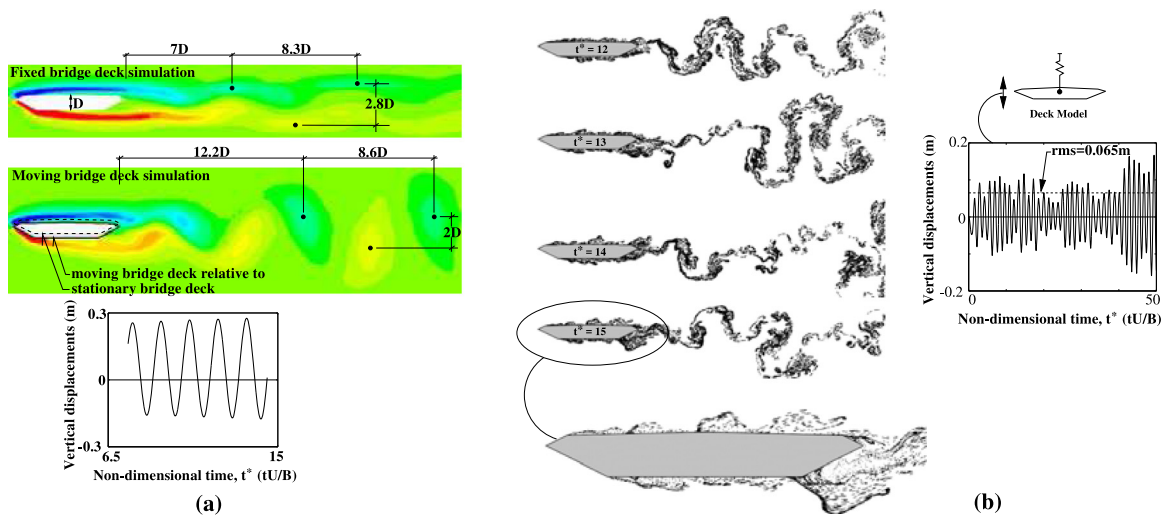


Fig. 4. Dynamic vortex evolutions around the vibrating bridge girder of Storebælt suspension bridge at the Reynolds number of 1.65×10^7 . (a) The finite element analysis based results and (b) the discrete vortex method based simulation results (rms, root-mean-square values) (Frandsen, 2000).

FEM results agreed well with the field monitoring results, as illustrated by Frandsen (2001b). Fig. 4(a) compares the wake vortex shedding from the steady and free-moving girder at a VIV wind velocity respectively where two arrays of counter-rotating shedding vortices were generated behind the girder. It should be noted that compared to the stationary girder, the wake width of the oscillating girder was wider due to the oscillation improving the periodicity of vortex shedding in its wake, as illustrated in Fig. 4 (a). In addition, for the DVM-based method results in Fig. 4(b), the flow separation occurred at the leading edge of the girder, convected downstream along the girder surface and alternately shed in the wake. It should be noted that the definition of leading edge here is from the structural perspective, instead of the aerodynamic point of view.

Besides, Chen et al. (2019) experimentally studied the vortex evolution around a 1:60-scaled girder model of the Storebælt suspension bridge. The VIV responses were recorded by two laser displacement sensors, and the flow changes in the wake were obtained by high-speed Particle Image Velocimetry (PIV) system. The shedding process of trailing vortices was shown in Fig. 5, where a typical 2S shedding mode was presented. In one period of VIV, the upper vortex shed at the T/4 phase (where T denotes the period per cycle), when the girder model reached the lowest point of the structural

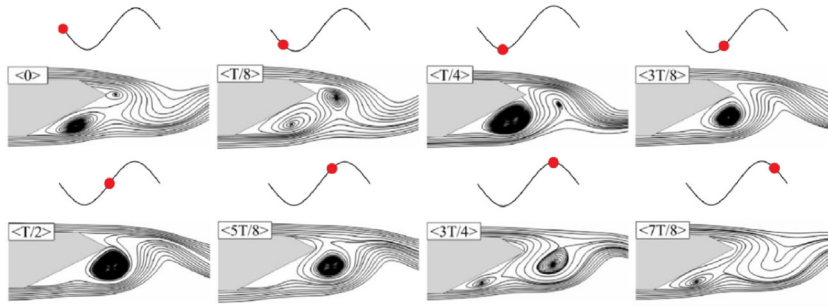


Fig. 5. Vortex shedding in the wake of a bared box girder of Storebælt suspension bridge at eight different phase in one VIV period (where T denotes the period per cycle). (Chen et al., 2019).



Fig. 6. Overview of Xihoumen suspension bridge (Li et al., 2014b).

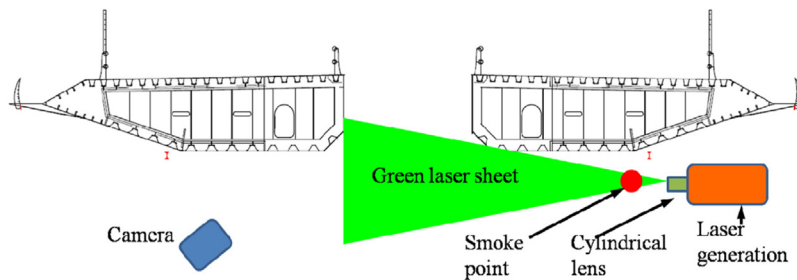


Fig. 7. Setup of on-site flow visualization for the real bridge girder (Li et al., 2014b).

responses. The lower vortex shed under the inclined side panel at the $3T/4$ phase, i.e. at the highest point of the sinusoidal VIV circle, which is consisted with [Komatsu and Kobayashi \(1980\)](#).

To further investigate the triggering mechanisms of VIV response in Storebælt bridge, [Terres-Nicoli and Kopp \(2009\)](#) conducted simultaneous pressure measurements to study the pressure distribution around the stationary and oscillating test model. Experimental results demonstrated that notable pressure fluctuations are observed at the trailing edge of stationary girder and the leading edge of oscillating girder, respectively, indicating that the TEVS are responsible for the onset of initial VIV which results in the impinging-shear-layer instability, and the aerodynamic interaction between LEV and TEV formed by motion induced instability gives cause for the strengthening of VIV amplitude.

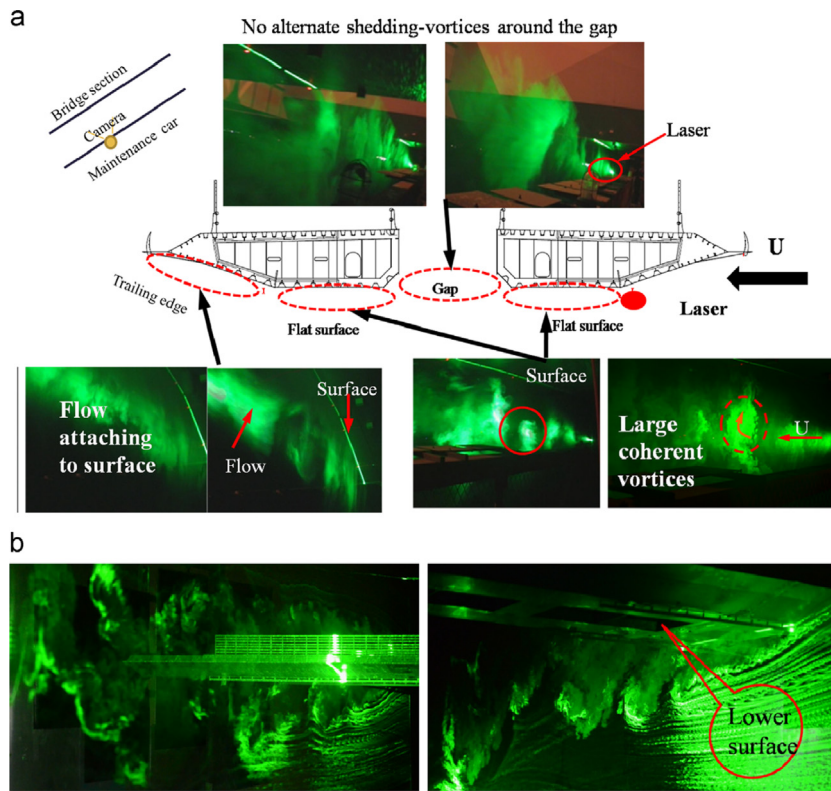


Fig. 8. Smoke visualization of the flow topologies around the lower surface of the separated girder. (a) full scale and (b) 1:25 test model in the wind tunnel (Li et al., 2014b).

2.2. Xihoumen suspension bridge: Twin-box girder

With the increasing requirement for a longer main-span and wider traffic lane, twin-box girders have been frequently adopted to improve the flutter instability of the long-span bridges. However, the twin-box girder is susceptible to VIV because of the gap vortex between the upstream and downstream girder (Li et al., 2011a). The Xihoumen suspension bridge is with a main span of 1650 m, connecting Zhoushan Archipelago with Ningbo, China, as shown in Fig. 6. It is the second-longest suspension bridge in the world (after Akashi Kaikyo bridge with the main span of 1991 m, Japan) when it was completed and opened to public in 2009 (Chen et al., 2018). A refined structural health monitoring (SHM) system consisting of several sensors for monitoring the incoming winds, surface pressures, structural vibrations and local surrounding flow patterns was installed on this bridge to monitor the wind environment and wind-induced vibrations (Li et al., 2014b). The in-situ monitoring results indicated that the inhomogeneity of spanwise flow characteristics plays a key role in the VIV of a full-scale bridge. Compared with 1:25 section model, the full-scale girder undergoes a higher amplitude of vertical VIV but no torsional VIV response.

To investigate the flow field on the lower surface, a field visualization, composed of three components: a smoke generator, a laser light sheet device and a CCD camera, was performed on the full-scale bridge as shown in Fig. 7. Fig. 8(a) shows the smoke visualization results (no VIV occurred) under a mean wind speed of 10.03 m/s. The corresponding wind direction was approximately perpendicular to the bridge axis, and the turbulence intensity I_u was 6.60%. It shows that the flow separated from the maintenance rail and reattached on the inclined plate of leeward box. Moreover, the separated vortex structures directly traveled across the gap. To further investigate the flow field around the twin-box girder, a flow visualization for the 1:25 section model was proposed as shown in Fig. 8(b), where the flow separated from the maintenance rail do not reattach on the lower surface of the section model, indicating that the wind tunnel test overestimate the flow separation at the trailing edge of twin-box girder, i.e., the flow field around the twin-box girder is substantially affected by Reynolds number (Re).

With regard to the Re effects, Schewe and Larsen (1998) tested the sectional models of the main girder in wind tunnel to cover Re of $10^4 \leq Re \leq 10^7$, and concluded that the VIV shows a significant dependence on the Re due to alterations of the topological flow structures. By employing a 3D large eddy simulation (3D-LES) method, Laima et al. (2018a) numerically investigated the Re effects on flow topologies around the main twin-box girder of the Xihoumen suspension bridge. Simulative results are described in Fig. 9, which shows that the flow patterns of the flow in the gap and wake region

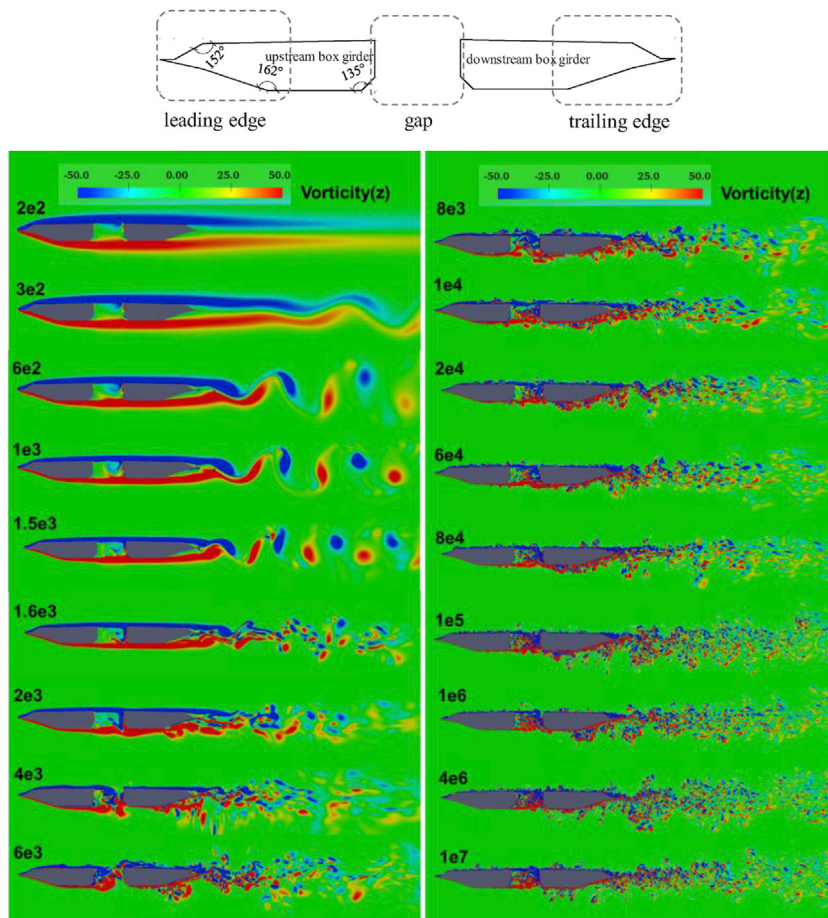


Fig. 9. Instantaneous vorticity contours around the bridge girder for various Reynolds numbers (Laima et al., 2018a).

suffer from the profound Re effect. With the increase of Re , the laminar flow around the box gradually transiting into the turbulence, and three-dimensional features are gradually formed in the wake. Moreover, Li et al. (2014a) experimentally investigated the Re effects on the VIV of a twin-box girder model, and found that a higher Re may lead to a wider lock-in range and a larger vibration amplitude.

There are two types of vortex shedding modes around the twin-box girder, i.e., 2S mode and 2(2S) mode, corresponding to one and two pairs of vortices shed in a vibration cycle. The 2S mode roots in the synchronization of vortex shedding and natural frequency, and the 2(2S) mode is caused by the 2:1 phase-locking of vortex shedding and model motion, resulting in two intervals of VIV. The gap vortex shedding during the first interval of vertical VIV is illustrated in Fig. 10, showing the distributions of swirling strength and streamlines at eight phases of a cycle (Yuan et al., 2017). The shear layers separate and roll up to form vortices in the gap region, which leads to the occurrence of the first VIV interval. The clockwise upper vortex is formed at the trailing corner of the upstream girder at phase (i) when the model moves upward beyond the equilibrium position. Then the vortex expands and moves downstream along the girder surface and sheds from the trailing edge of the downstream girder. The vortex shedding in the gap keeps same phase with that in the wake, and both are in the classical 2S mode.

In terms of the sharp-edged bluff body, the vortex excitation can be mainly divided into two categories: one-and two-shear layer instability (Matsumoto, 1999). Nakashima et al. (2010) mounted a splitter plate behind a bluff prism with aspect ratio ranging from 2.0 to 5.0, which is an effective way to prevent the interaction of the upper and lower separated vortex. Experimental results demonstrated that the VIV occurred even though the splitter plates were installed, indicating that both two-shear and one-shear layer instability play a key role in the VIV of the bluff prism. It is generally accepted that the separated shear layer emitted from the leading edge of a girder often becomes unstable to roll up in discrete vortices on the surface of the downstream corner. Such a flow instability, which pertains to the more universal category of flow instabilities correlated with jets and wakes, is described as the impinging-shear-layer instability (Rockwell and Naudascher, 1978; Hu et al., 2018). Therefore, the existence of attachment components such as handrails, maintenance rails, and crash barriers, can provide an appropriate modification on the girder geometry to suppress the impinging-shear layer instability and then eliminate the VIV induced by motion-induced vortices. Nagao et al. (1997) discussed the

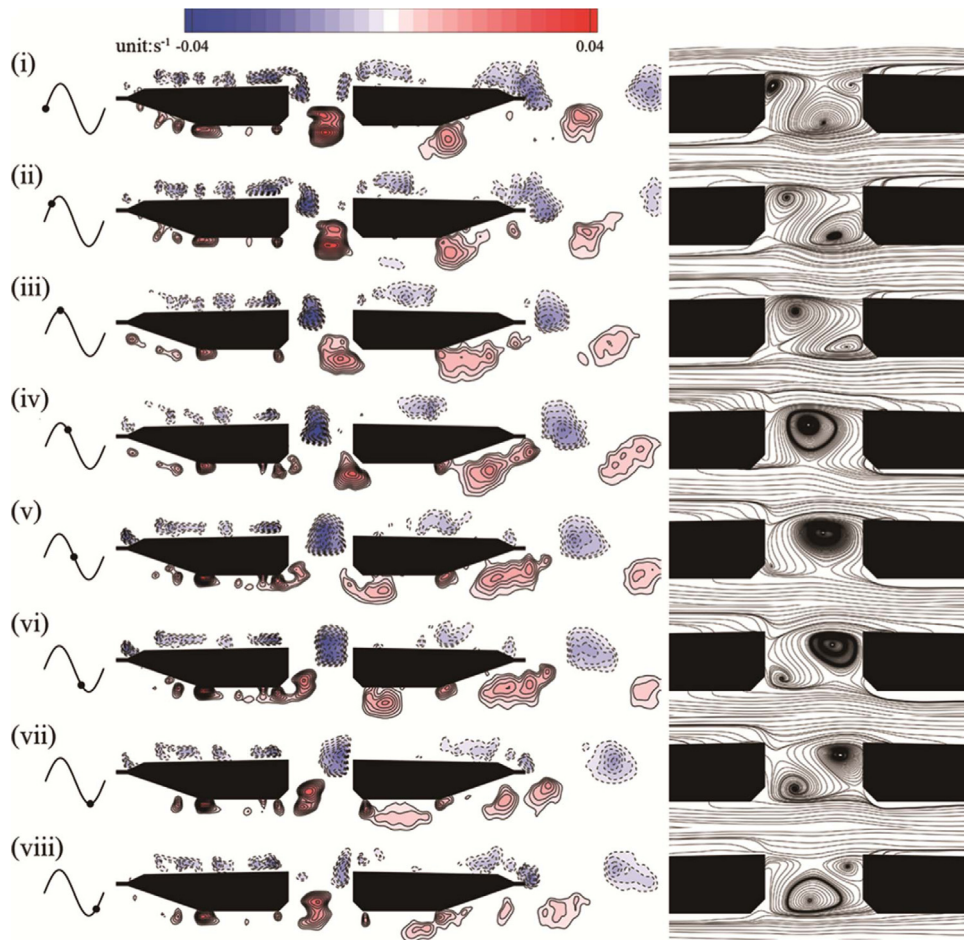


Fig. 10. Dynamic flow evolutions around the separated twin-box girder in a period of vertical VIV. The left panel is the global flow, while the left is the gap flow (Yuan et al., 2017).

influence of the handrail on the VIV of a girder with an aspect ratio of 5.77. In their experimental research, the position and dimension of the horizontal bar have significant influence on suppressing the torsional VIV due to the interaction between the separated shear layer and the horizontal bar. However, as the first separated flow impinges the horizontal bar and second separated flow from its bottom corner, the fluctuating pressure amplitude is amplified and the VIV response is increased. Laima et al. (2018b) investigated the flow structures around the twin-box girder with the attachments via wind-tunnel experiments and found that the vortex evolutions around the separated girder with and without attachments are obviously different, as shown in Figs. 11 to 13. For the girder without attachments in Fig. 11, the flow separates and reattaches at the windward corner of the upstream girder. A series of Kelvin–Helmholtz(K–H) vortices generated from the shear layer convect downstream and directly travel across the gap. Finally the vortices shed at the trailing edge of the downstream girder. In virtue of a streamlined shape, the flow separation at the leading edge is moderate and the width of the wake is narrow. However, for the configuration of attachments, the flow significantly separates near the leading attachments, leading to the larger spacing of two shear layers shed from upper and lower sides of the windward girder, according to the decreasing width of the equivalent gap ratio. That contributes to avoiding the gap vortex shedding, in this way the VIV excited by the gap vortex formation may be suppressed.

2.3. Humen suspension bridge: A more recent case

On May 5, 2020, Humen suspension bridge (Guangzhou, China, its cross-section of the main girder is shown in Fig. 14) was observed to vibrate notably due to the action of moderate wind. The vibration started at 14:30 and continued intermittently for more than two hours. On the basis of eye-observation, the vibration was mainly dominated by the vertical vibration mode, and no torsional mode vibration was observed. The maximal amplitude was roughly estimated to be 0.3 m. The vibration was initially judged as VIV. At 15:32, the traffic was announced closed in response to public

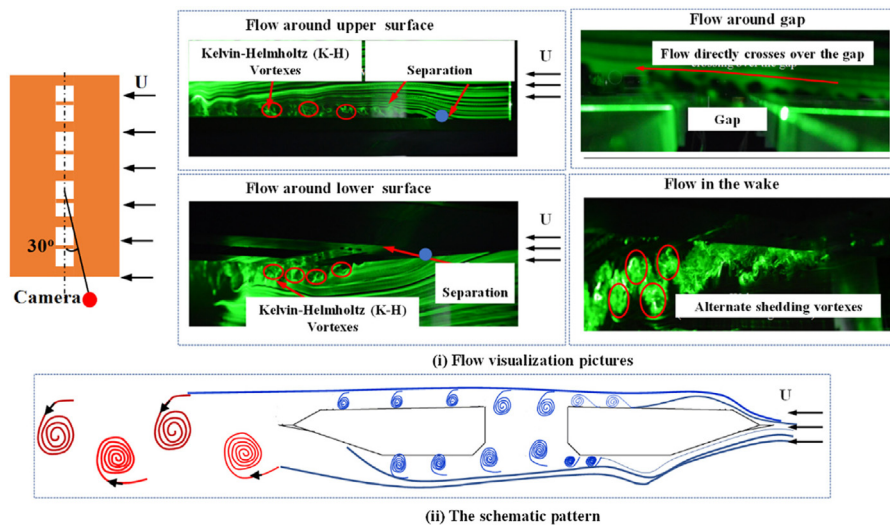


Fig. 11. Flow structures around the natural separated girder without any attachments. The flow is visualized by smoke (Laima et al., 2018b).

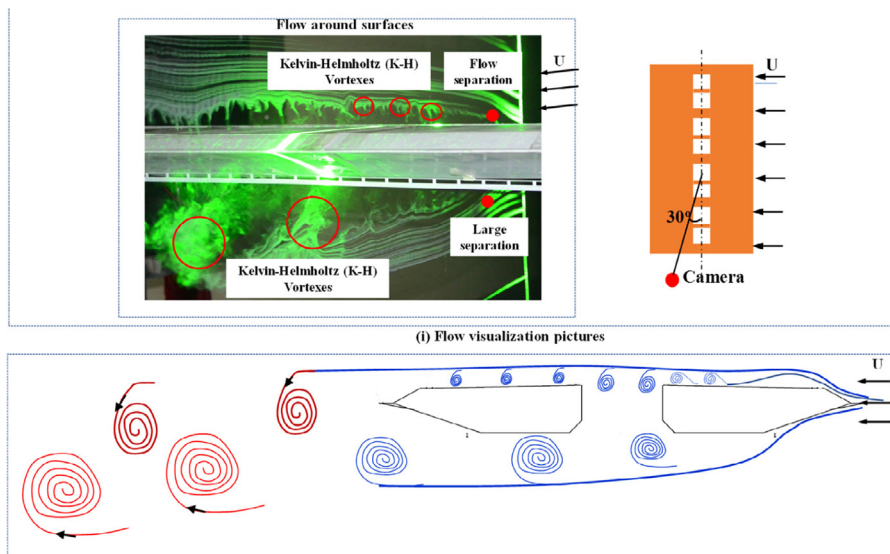


Fig. 12. Flow structures around the separated girder with maintenance rails (Laima et al., 2018b).

concerns. When preparing the following time-lines of the VIV event of Humen suspension bridge, we have referred to the Wechat public account of “Tongji Wind Engineering”.

At the occurrence of VIV, the vertical hangers of the bridge were being replaced. Isolation facilities, which were generally referred to as water horses, were temporarily installed on both sides of the bridge deck for the sake of traffic safety. It should be noted that the height of water horses was 0.8 m. Thus, the VIV of the Humen suspension bridge was judged to be caused by the temporary installation of water horses during the regular maintenance process. The water horses had changed notably the aerodynamic shape of the closed steel box girder. It was recommended to remove the water horses immediately to get rid of VIV. Then, all the water horses were dismantled swiftly. However, the bridge girder still kept moving, indicating that the VIV was not completely caused by the water horses.

On May 7th, the displacement response of Humen Bridge and the wind environment of the bridge deck had a chance to be precisely measured and investigated, by the use of video imaging displacement sensors and portable anemometers. The wind speed was measured to be 8–10 m/s, with attack angles of 90 ± 20 degrees off to the main girder. The maximal unilateral amplitude of VIV was 17.5 cm. According to the on-site investigations, when the water horses were arranged, the vibration frequency at 8–10 m s^{-1} wind speed was about 0.368 Hz, while the vibration frequency was 0.225 Hz after the water horses were removed. It indicated that the bridge girder's Strouhal number changed significantly before and after

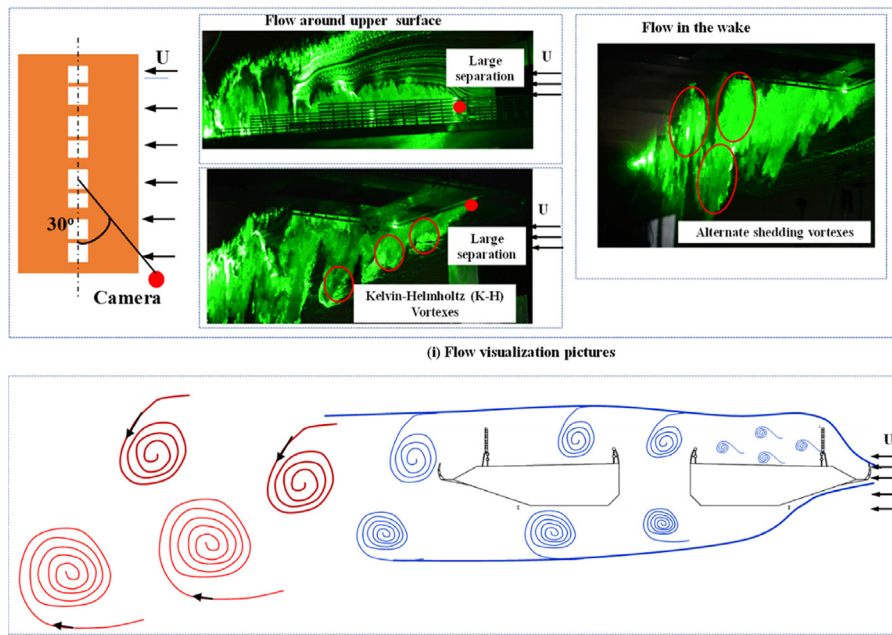


Fig. 13. Flow structures around the separated girder with maintenance rails, wind barriers, handrails and crash barriers (Laima et al., 2018b).

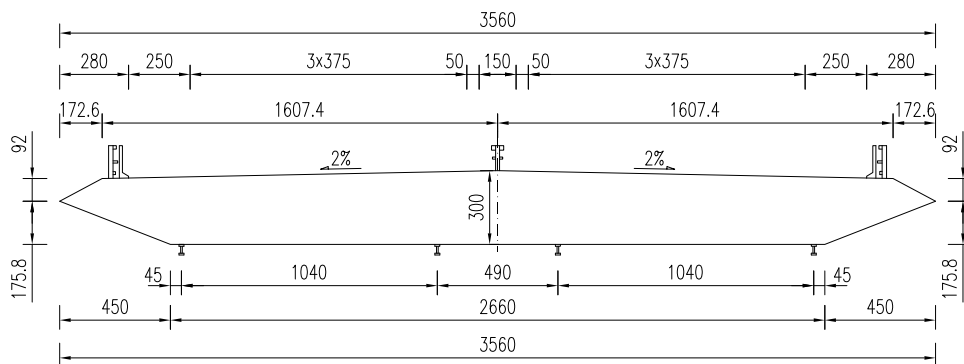


Fig. 14. Schematic cross-section of the main girder of Humen suspension bridge (unit:cm).

the removal of water horses. Besides, the VIV exhibit mainly three vibration modes: third-order symmetrically vertical bending mode, second-order symmetrically vertical bending mode and second-order asymmetrically vertical bending mode. Based on in-situ monitoring results, it was proposed that the Humen Bridge experienced two VIV events. The first one was caused by the water horses. However, the cause of the second VIV event should be carefully investigated.

In terms of CFD simulations, the flow around the main girder of Humen suspension Bridge is investigated. As the water horses were placed near the upstream railing of the main girder, a regularly periodic vortex shedding process was observed on the main girder. The vortex moved downstream along the main girder and decoupled from the Von Karman vortex at the tail of the main girder. It was shown that the frequency of the vortex shedding matched well with the bridge vibration. Thus, it was confirmed that the vibration was VIV. In addition, as the water horses were removed, it was noticed that when the incoming flow passed through the outer maintenance track positioned at the lower surface of leading edge, it would also produce a strong periodic vortex shedding from the girder and cause VIV. Also, the lidar detected periodic fluctuations with a frequency of 0.273 Hz in the downstream wake of the main girder of the bridge, after the removal of water horses.

Another finding of in-situ monitoring was that, the structural damping ratio of Humen Bridge had decreased significantly (estimated up to 50%). Therefore, adding tuned mass dampers (TMD) became the first choice for damping augment and VIV suppression. However, the implementation process of TMD required a certain period of time, and aerodynamic countermeasures were considered for temporary use. Therefore, installing barrier spoilers on the top of barriers to control the vortex shedding from the upper surface of the main girder; and the outer maintenance rails were suggested be

removed to eliminate the vortex shedding at the bottom. After barrier spoilers were installed along the entire bridge on May 13th, the maximum amplitude of VIV was decreased to half. Meanwhile, it was demonstrated by lidar that no regularly periodic vortex shedding was detected behind the bridge. It should be noted that the barrier spoilers were temporary countermeasures, and for long-term safety, the outer maintenance rails need to be removed to eliminate the generation of regular vortices at the bottom of the main girder.

After 10 closing days of public traffic, Humen bridge was re-opened to public at 9:00 am on May 15th. Nevertheless, detailed studies, including scale model-based wind tunnel tests, LES-based simulations, of the VIV mechanism of Humen Bridge are still ongoing and more scientific results are about to be reported in the near future.

2.4. Summary

In this section, the vortex dynamics-based mechanism of two representative types of bridges (i.e. single-box girder and twin-box girder) is detailedly expounded, which provides a new perspective for the VIV control in bridge engineering. Besides, a recent VIV case of Humen suspension bridge is introduced briefly. Compared to structural countermeasures (i.e. increasing its frequency by setting intermediate supports) and mechanical countermeasures (e.g., tuned mass dampers), the countermeasures based on the vortex dynamics impose disturbances into the vortex shedding to avoid the development of VIV in a simple and reliable way. Based on whether external energy input is needed, the flow control strategies can be divided into passive and active ones, which are described in detail in following two sections.

3. Passive flow control

Passive control strategies are conducted to attain a better aerodynamic performance of a bluff body by optimizing the geometric shape or interfering the surrounding flow field. The passive control techniques have been widely applied in engineering practice because of its easy, flexible, and energy-efficient implementation. In this section, the latest successful methods based on vortex dynamics for suppressing the VIV of the girder are presented, including geometry optimization, attachments, and self-issued jets.

3.1. Geometry optimization

For the bridge, its aerodynamic and aeroelastic instability is primarily related to the shape of the girder, therefore the geometry optimization for the long-span bridge is one of the most important steps at the designing phase. Adjusting the inclined panel, wind fairing and gap ratio are verified to be a successful approach to optimize aerodynamic performance of a bridge girder.

Most modern bridges adopt a semi-streamlined girder for better aeroelastic instability. Although the semi-streamlined girder has the advantages of low wind loading and high critical flutter speed, it may suffer from the VIV at a low wind velocity. By optimizing the geometrical shape of the girder, the flow separation could be attenuated and the vortical distribution in the near wake might be changed, leading to a great suppression of the VIV amplitude. [Larsen and Wall \(2012\)](#) investigated the influence of the angle between the inclined panel and bottom plate on the VIV of the girder via wind tunnel experiments, and concluded that for the Storebælt suspension bridge (and the Halogaland suspension bridge), 15 deg is the optimum degree for preventing the flow separation at the trailing edge. In the VIV state, two contra-rotating vortices formed on both sides near the nose-tail line of the girder provide enough momentum to magnify the VIV amplitude, where the positive vortex below the tail is larger than the negative one above ([Wang et al., 2011](#)). Therefore, the periodical vortex formation under the inclined web (i.e., the large-scale positive vortex below the tail) is the main origin of VIV. When the angle of inclined web is adjusted to 15 deg, a small negative pressure area below the tail was formed in the wake and the flow near the lower surface was mainly attached to the inclined web, which makes it hard to form a large-scale vortex. In this way, the vortex formation under the inclined web is diminished and the large-scale coherent vortical structures are formed further downstream, as thus the VIV is effectively suppressed ([Larsen and Wall, 2012](#)).

The angle of wind fairing (β) also plays a major role in the aerodynamic performance of a streamlined girder. [Haque et al. \(2016\)](#) carried out an unsteady RANS simulation to explore the control effect of various angles of the wind fairing by adjusting the upper and lower plate slope on the aeroelastic stability. It was found that the aerodynamic forces exerting on the test model are reduced with the decline of plate slopes, and the flow separations on the leading and trailing edge are susceptible to the fairing shape. Among them, the trailing flow separation primarily controls the vortex shedding patterns as well as the fluctuating aerodynamic loads, and the leading flow separation mainly influences the mean aerodynamic forces and Strouhal number. [He et al. \(2017\)](#) experimentally studied the effect of wind fairing angle and the location of the wind fairing on the aerodynamic performance of a streamlined girder. It shows that the aeroelastic behaviors are highly sensitive to these geometrical parameters when the angle of attack exceeds a threshold value, which alerts TEVS to ILEV. Besides, a higher wind fairing nose and a larger wind fairing angle were favorable in suppressing the VIV response. In addition, [Wang et al. \(2011\)](#) experimentally investigated the effect of variable height and angle of wind fairing on the VIV of a bridge girder. They found that a blunter wind fairing promotes the flow separation on the girder, where the vortices were formed and convect downstream, which may enhance the VIV response. For a low slope of the lower

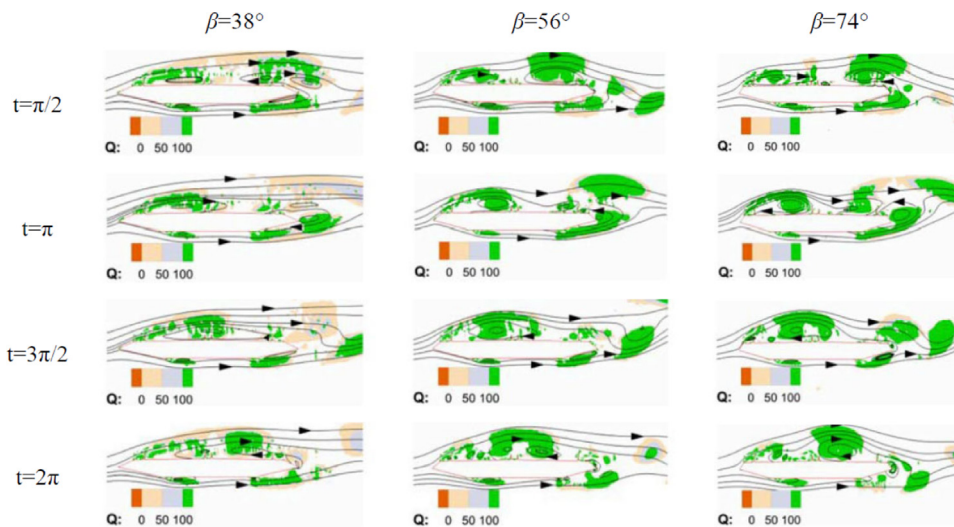


Fig. 15. The contours of vorticity magnitude and streamlines with different wind fairing angle in a period (Li et al., 2018).

inclined panel, the increase of the width of wind fairing may cause the generation of large-scale vortices, which has negative effects on the aeroelastic stability. Zhou et al. (2015b) combined velocity field measurements and CFD numerical simulations to investigate the influence of three types of wind fairings on VIV stabilization, i.e., trapezoidal, airfoil and smaller airfoil wind fairings, and found that all the three countermeasures can mitigate VIV to some extent. Recently, Li et al. (2018) simulated the aeroelastic response of a streamlined girder with different fairing angles. The simulation results suggested that the fairing angle has more influence on the upper surface rather than the lower surface, because variations of the fairing angle can lead to an obviously change of upper plate slope. Furthermore, with the decrement of wind fairing angle, the VIV amplitude is effectively reduced and the width of vortices is narrowed as shown in Fig. 15. These vortices move downstream, leading to the alteration of pressure distributions on the girder surface in a period. As the fairing angle increases, the pressure difference between the upper and lower surface is amplified, and then enhance VIV response. Therefore, preventing the formation of large vortex at upper leading edge plays a significant role in improving the aerodynamic performance of a streamlined box girder.

For the twin-box girder, Chen et al. (2014b) conducted experimental investigation to analyze the characteristics of the flow field around a steady twin-box girder models with gap ratios (G/D , where G is the gap width, and D is the girder height) varied from 0.85 to 3.40, as illustrated in Fig. 16. The unsteady vortical structures are mainly generated from the trailing edge of the downstream box with small gap ratio (i.e., $G/D = 0.85$) as shown in Fig. 16(a), where the vortices shed predominately in the wake. Whereas the gap ratio increases beyond 1.70 as illustrated in Fig. 16(b)–(d), the gap vortex occurs and turns to be stronger with the gap ratio, and the vortex shedding can be observed in the gap and wake region. In those case, the gap vortices impact on the leading edge of the leeward box, leading to high pressure fluctuations on the downstream box, which is consistent with (Larsen et al., 2008). Furthermore, Laima and Li (2015) experimentally investigated the influence of gap ratio (G/D) on the flow features and VIV responses of a twin-box girder, and concluded that the flow topology can be divided into three categories: Pattern A, Pattern B and Pattern C depending on different gap ratio, as illustrated in Fig. 17. For pattern A, the periodically shedding vortex occurs on the leeward box girder and the pressure fluctuations of the sectional model are slight at a small gap ratio. As the gap ratio locates in the range of Pattern B, the shear layers separated from the trailing edge of the windward box periodically develop into Kármán vortices in the gap, which move downstream and impact on the leading edge of the leeward box, leading to flow-separation at the leading edge corners. In Pattern C, the gap width is large and no flow interaction between the flow field around the windward and leeward box, which can be seen as two independent girders. Moreover, four types of the VIV response may occur at different gap ratio, including no VIV ($G/D \leq 1.710$), vertical VIV ($1.710 \leq G/D \leq 2.138$), vertical and torsional VIV ($2.566 \leq G/D \leq 2.933$) and torsional VIV ($G/D \geq 3.421$). When $G/D = 1.71$, the inchoative stage of vertical VIV is caused by the alternating vortex shedding process in the wake of the leeward box, and then the motion-induced vortices formed in the gap are responsible for the gradually amplified vibration. While for $G/D \geq 2.138$, the torsional VIV is mainly caused by the alternating vortex shedding process in the gap, and the vibration trace is irregular duo to the effect of turbulence on leeward box.

3.2. Attachments

Installing attachments on the main girder is a normal approach to mitigate the VIV, for instance, spoilers, guide vanes, failing plates (Fujino and Siringoringo, 2013; Zhou et al., 2015a). The Guide vanes (Larsen, 1993; Larsen and Poulin, 2005),

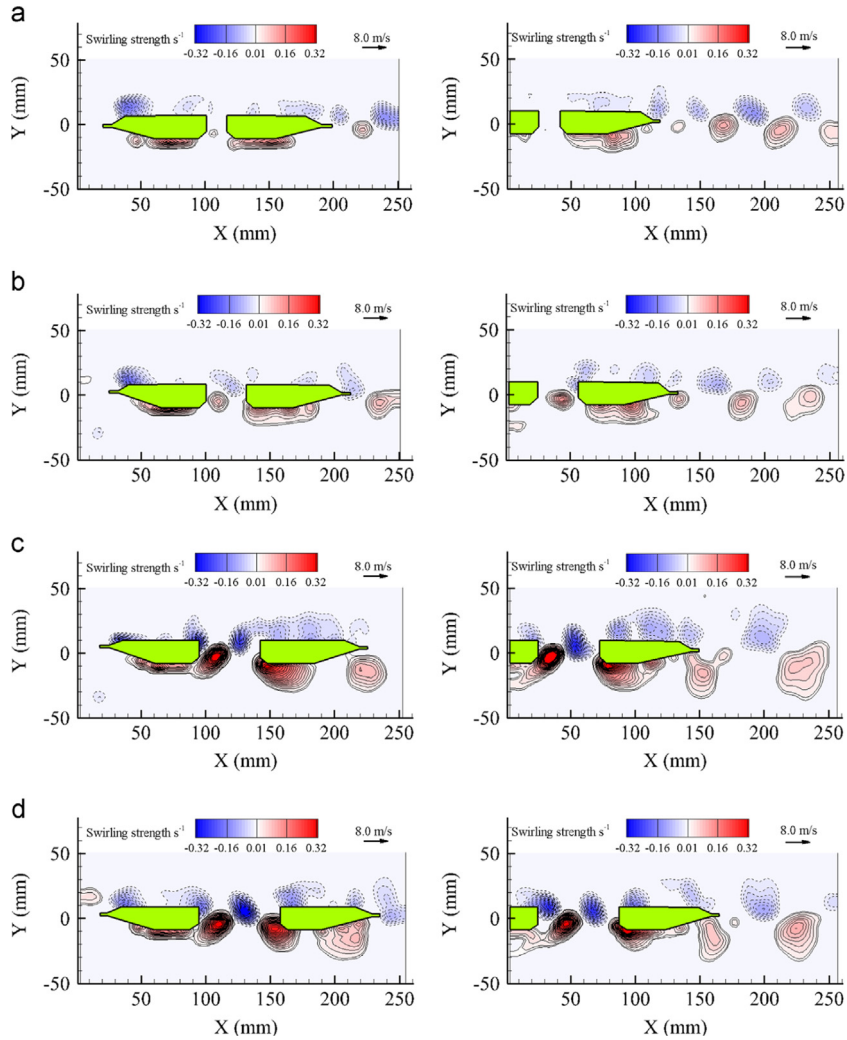


Fig. 16. Instantaneous vorticity evolutions around the test model with different gap ratios. From (a) to (d), gap ratio is 0.85, 1.70, 2.55 and 3.40 respectively (Chen et al., 2014b).

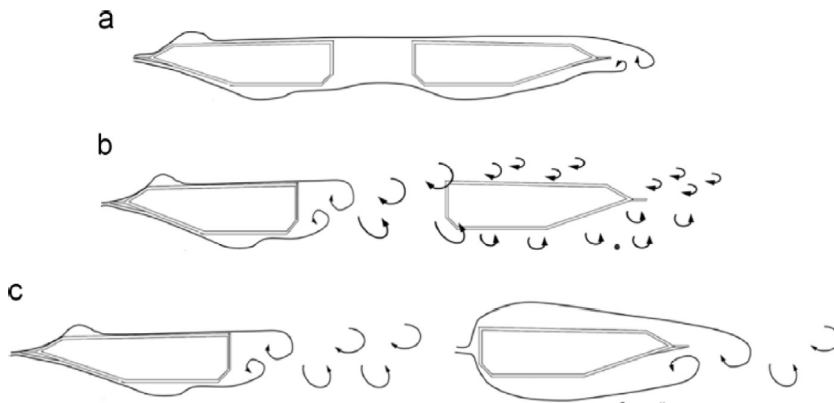


Fig. 17. The swirling strength distribution of twin-box girder with different gap ratios. According to the vortical characteristics near the gap, three types of flow topology could be concluded (Laima and Li, 2015).

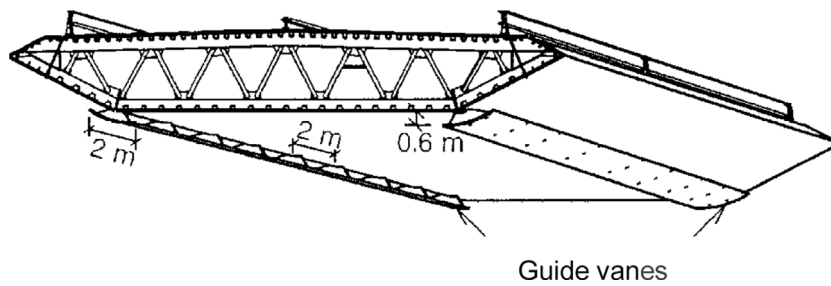


Fig. 18. Experimental arrangements of guide vanes mounted along the lower side panel joint of the main girder on the sectional test model (Larsen et al., 2000).

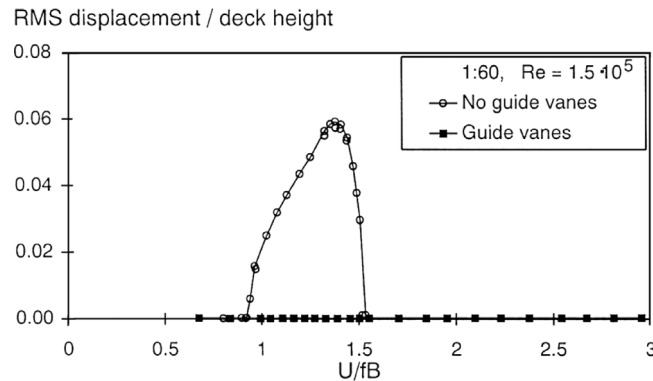


Fig. 19. VIV response of the sectional model with and without guide vanes (Larsen et al., 2000).

as illustrated in Fig. 18, can guide the oncoming flow into the wake region to push the Kármán vortex downstream and then reduce the fluctuating pressures on the lower inclined panel (Larsen and Wall, 2012), in this way the VIV responses of the girder is eliminated as shown in Fig. 19, where the reduced wind velocity $Ur = U/fsB$, B is the overall deck width. It is noteworthy that its control effect on VIV is influenced by Re . At low Re , the boundary layer would be too thicker to prevent sufficient flow passing through the guide vanes, which may amplify the VIV responses (Larsen et al., 2008).

Dale (2004) investigated the influence of baffle plates on the VIV response of a bridge via wind tunnel tests, and found that the baffle plates can effectively suppress the flow separation and then VIV. Nevertheless, the optimal position of baffle plates for attenuating the torsional VIV was different from that of the vertical VIV. Sarwar and Ishihara (2010) numerically investigated the VIV responses of a box girder with double flaps on the windward and leeward edge and found that the presence of double flaps improved the negative aerodynamic damping slightly, and eliminates the vortex shedding on the upper surface, in this way the VIV is suppressed. Sakai et al. (1993) proposed an easy and effective aerodynamic appendage called tip plate, which successfully manipulate the torsional VIV by disturbing the vortex formation procedure at the leading edge of the girder. Wang et al. (2011) experimentally investigated the different types of guide wings on the girder edge as described in Fig. 20 and the correlation between the guide wing and VIV response was analyzed. In their experiments, the aerodynamic stability was apparently improved along with the increment in the width of the guide wing at a positive inclination because the guide wing can smooth the fluid structures around the sectional model. However, Li et al. (2011b) pointed out that the guide wing mainly has good performance on suppressing the vertical VIV, while for torsional VIV the control effect is opposite.

In terms of twin-box girder, many passive attachments were proposed as the twin-box girder is more susceptible to VIV than the single-box girder. According to the field monitor results of the twin-box girder (Li et al., 2011a), the shedding vortex only occurs in the central gap and the tail region of the leeward girder in the early stage of VIV. With the development of the VIV, the vortex shedding was gradually enhanced, and spread to the whole bottom surfaces of the leeward girder and the trailing edge of the windward girder, resulting in the connection of the vortex shedding regions between the central gap and bottom surface in the lock-in stage. Therefore, the central gap was a key affecting the VIV response of a twin-box girder. Ge et al. (2009) carried out an experimental study to evaluate the aerodynamic performance of a twin-box girder installed with vertical central stabilizer (VCS). It was demonstrated that the VCS can effectively augment the critical wind speed, and then suppress the VIV response. On this basis, Zhou et al. (2018) comprehensively assessed the effects of critical parameters of VCS on the VIV of a twin-box girder in Fig. 21 and point out that the height of VCS and slot width ratio are the key aspects: a higher VCS may cause the heaving VIV to occur. Laima et al. (2013) and Yang et al. (2016) covered the grid plates on the central gap of the twin-box girder to avoid the generation of the gap

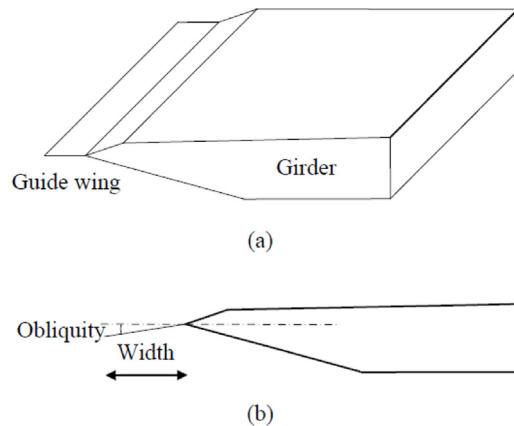


Fig. 20. Axonometrical and elevation views of the girder configuration modified with guide wing (Wang et al., 2011).

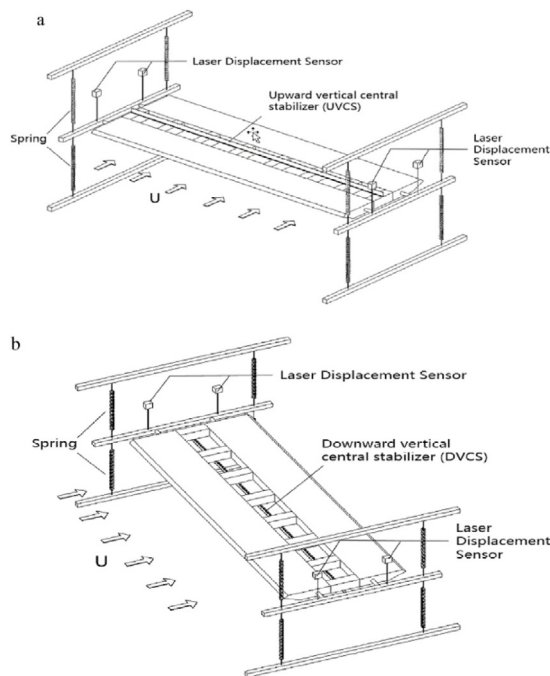


Fig. 21. Configuration of elastically suspended models modified with: (a) upward vertical central stabilizer; (b) downward vertical central stabilizer (Zhou et al., 2018).

vortex, as shown in Fig. 22. It seems that the optimum grid ratio is 20% for mitigating the heaving VIV by the grid plates, and the grid ratio of 0% is the most beneficial grid porosity to eliminate the VIV response both in vertical and torsional. However, the grid plates with a grid ratio of 0% may reduce the flutter critical velocity, therefore the grid porosity should be designed reasonably in consideration of both VIV and flutter. Zhou (2010) mounted horizontal plates on both sides of the central gap and found that as the width of the horizontal plate increases, the control effect on VIV shows a trend of increasing firstly and then decreasing. Moreover, the horizontal flow-isolating plate showed a better control performance on torsional VIV than vertical VIV.

Yang et al. (2014) mounted the guide vanes at the bottom surfaces near the central gap, namely ‘internal guide vanes’ to control the VIV of a twin-box girder at various AOA, as shown in Fig. 23. The results indicated that the VIV response is highly affected by the chamfers of the central gap. If the central gap of the main girder is chamfered, the internal guide vanes may have a better control effect on the torsional VIV; on the contrast, the control effect of vertical VIV is more significant. Larose et al. (2003) used a 1:20 sectional model in a wind tunnel to investigate the VIV of the main girder of Hong Kong Stonecutters bridge. Two torsional “lock-in” regions were noted in the wind tunnel test and only one “lock-in” region for the vertical mode of VIV was observed. It was also demonstrated that the guide vanes and maintenance rails

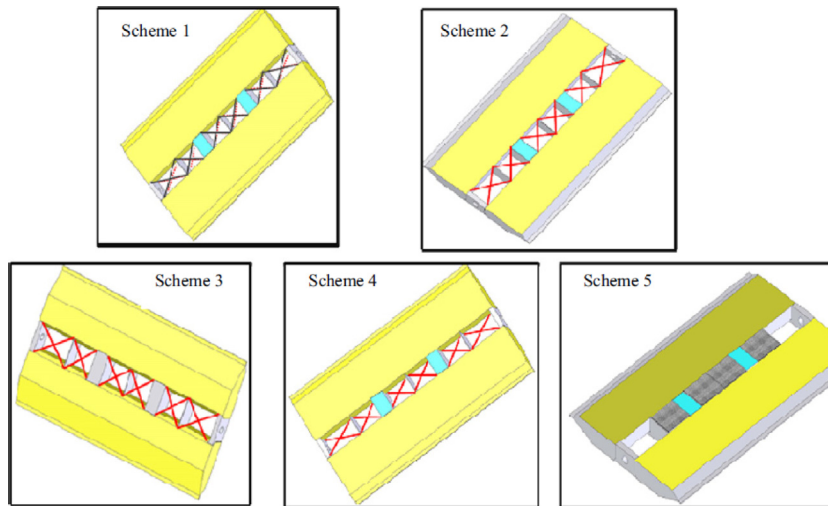


Fig. 22. Configuration of section models with five different kinds of grid plates: (a) cross bars on both sides of the central slot, (b) cross bars on the upper side of the central slot, (c) cross bars on the lower side of the central slot, (d) cross bars in the central slot, and (e) light flat plates on both sides of the central slot (Laima et al., 2013).

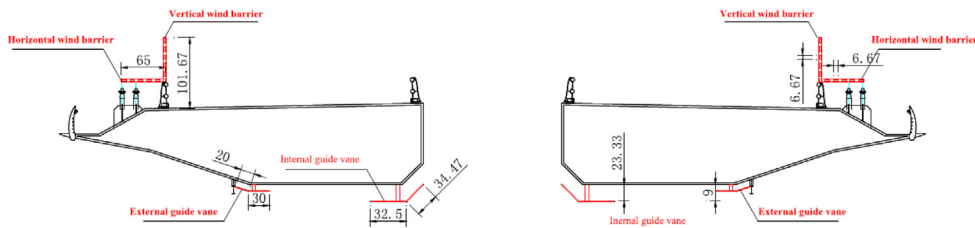


Fig. 23. Schematic of the twin-box girder modified with internal and external guide vanes (Yang et al., 2016).

were not so effective for VIV suppression at low Reynolds number but became much more efficient at high Reynolds numbers. The guide vanes were found to prevent the formation, shedding and drifting of vortices to impinge on the downwind box, and thus the root-mean-square response can be suppressed by nearly 40% (Hui and Yau, 2010).

Besides the control strategies aforementioned, which impose two-dimensional perturbations into the boundary layer or wake flow to control the VIV, another type of approach, i.e., three-dimensional forcing (3D forcing), is commonly applied in the engineering field of aeronautics and energy. According to the review of Kim and Choi (2005), the momentum needed for vortex shedding suppression from 3D disturbances is quite smaller than that from traditional 2D perturbations, indicating that the 3D forcing (i.e., spanwise varying distribution) shows better efficiency than 2D forcing (i.e., spanwise homogeneous distribution). The essence of 3D disturbances can be attributed to the motivation of the secondary wake instability, which is also referred to as 3D instability, caused by the instability of spanwise vortex (i.e., von Kármán street) in the spanwise direction in Fig. 24 (Hwang et al., 2013). Note that this spanwise variation would not result in out of phase for the vortex-induced forces due to its own symmetry. Under the proper geometrical modifications along the trailing edge of the structure, the quasi-synchronous shedding of the von Karman Vortices is rarely influenced, but the spanwise correlation is weakened (Doddipatla et al., 2017). Therefore, the von Karman vortices are suppressed and the aerodynamic characteristics of the structure are changed (Tombazis and Bearman, 1997; Doddipatla et al., 2017).

In general, there are mainly two types of 3D instability induced in the near wake of a bluff body, namely the Mode A (i.e., core instability) and Mode B (i.e., braid instability), on the basis of the Re (Williamson, 1996b,a). These two modes show as waves in the spanwise vortices, which are developed to the pairs of contrarotation streamwise vortices attaching the von Kármán streets. For Mode A as a elliptical instability, pairs of streamwise vortices alter the rotating direction twice in a period and the spanwise wavelength is three to five times the characteristic dimension of the bluff body. While for Mode B, as a hyperbolic instability, the streamwise vortex pairs keep the rotating direction and the wavelength of spanwise is approximately one time characteristic dimension (Williamson, 1996b,a) as illustrated in Figs. 25 and 26. These streamwise vortices may give rise to the dislocation of spanwise vortices and the energy transformation from the spanwise vortex to the streamwise vortex (Doddipatla et al., 2008).

The attachments periodically arranged on the girder along the axis direction can impose three-dimensional perturbations into the wake to improve the aerodynamic performance. El-Gammal et al. (2007) utilized the spanwise sinusoidal

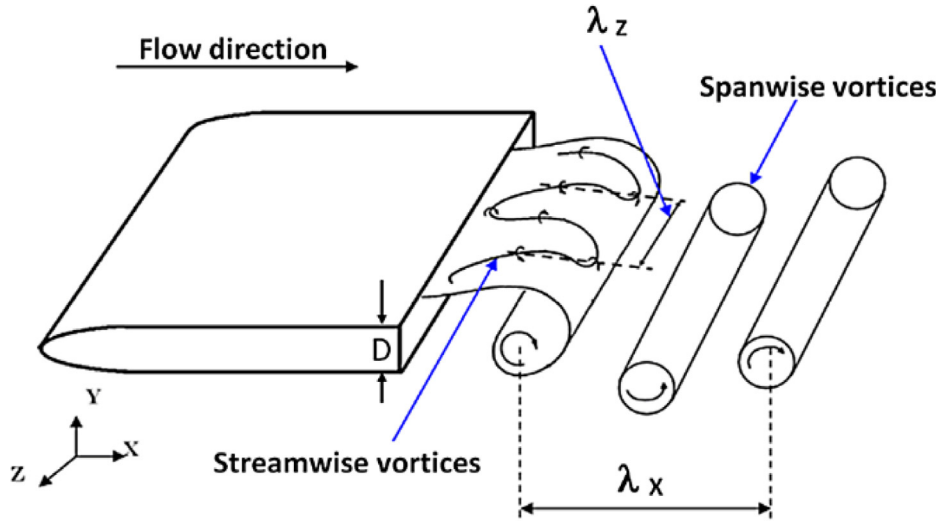


Fig. 24. Schematic of 3D instability (Doddipatla et al., 2017).

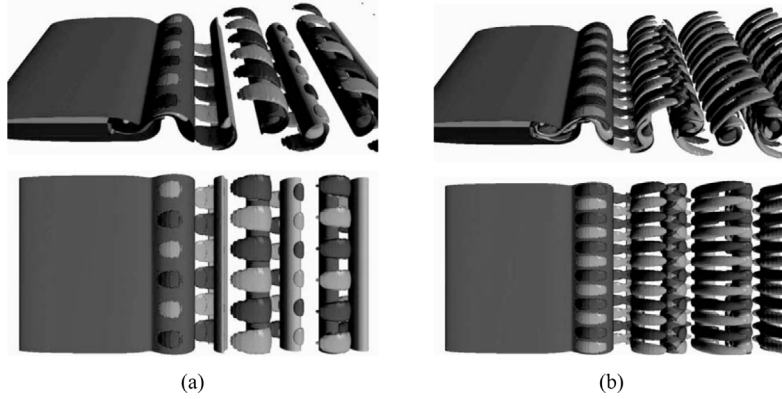


Fig. 25. Schematics of the vortical topologies of two types of 3D instability: (a) Mode A; (b) Mode B (Ryan et al., 2005).

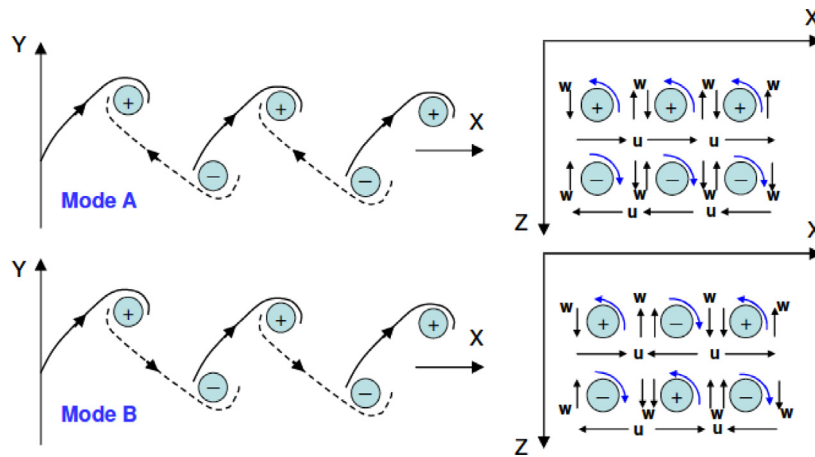


Fig. 26. Description of Mode A and Mode B. (The circle represents the spanwise vortex, and the curve represents the streamwise vortex. The sign of the rotating direction conforms to the right-hand rule, i.e., + means counterclockwise rotation, - means clockwise rotation; solid and dashed lines indicate different directions of rotation, which are also marked by arrows.) (Doddipatla et al., 2008).

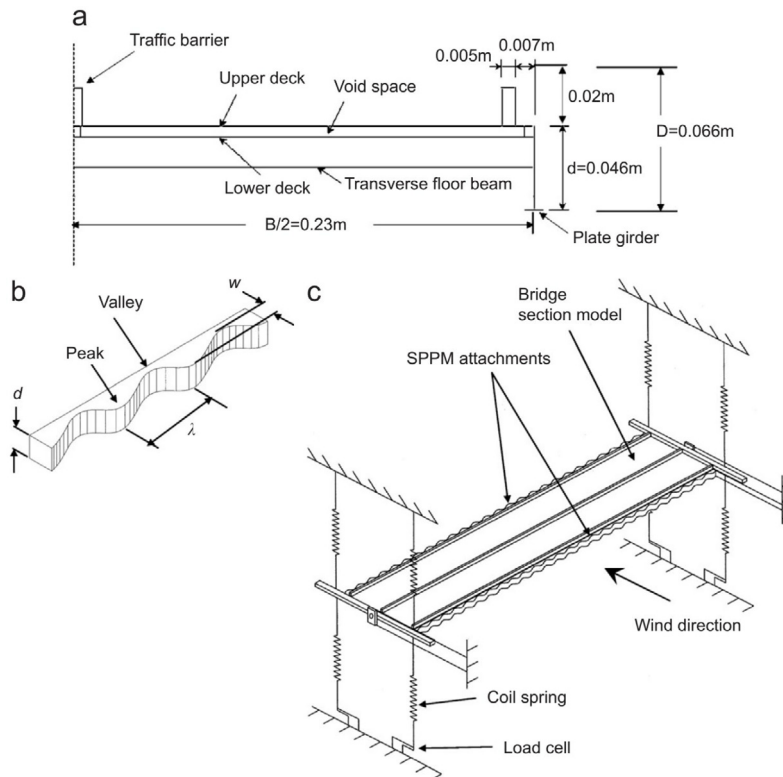


Fig. 27. Configuration of experimental setup: (a) intersecting surface of the baseline case; (b) microstructure of spanwise sinusoidal perturbation; (c) axonometric drawing of the test model modified with spanwise sinusoidal perturbation (El-Gammal et al., 2007).

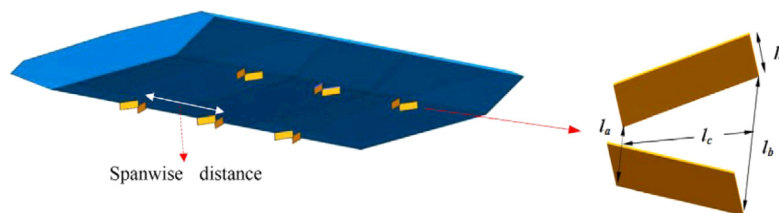


Fig. 28. Schematic of sectional test model attached by a number of passive vortex generators and its parameter (Xin et al., 2018a).

perturbation method (SPPM) on a plate girder as shown in Fig. 27, and found that the sinusoidal edge installed on two sides of the girder can dislocate the shedding vortex along the spanwise direction and suppress the vertical VIV. Furthermore, a higher wave steepness can attain better performance on stabilizing the attached flow, eliminating the shedding peak of the fluctuating pressure, weakening the strength of spanwise vortices and attenuating the VIV response. Zhang et al. (2019) introduced the vane vortex generators, which were widely used in aeronautics engineering, into the field of bridge engineering and arranged them on the lower surface of a single-box girder as illustrated in Fig. 28, and investigate the control effect on the shedding vortex. The results demonstrated that the lower shedding vortex is disturbed by the vortex generators, and the vortex generator can redistribute the energy spectrum, convert the energy from vortex shedding mode to other low-frequency modes, in this way the vortex-induced force is markedly suppressed. The Floquet stability analysis was applied to illustrate that the optimum effect occurs when the spanwise wavelength (λ/H) is 2, corresponding to the mode-A instability in the wake (Xin et al., 2018b). Furthermore, Xin et al. (2018a) experimentally investigated the control effect of the vane generator on VIV responses of a single-box girder. The PVGs can induce streamwise counter-rotating vortices, which can give rise to the dislocation of spanwise Kármán vortices. As a result, the spanwise shedding vortex was disturbed and the VIV responses were suppressed significantly.

To avoid the wind-induced vibration and improve the robustness of bridge system, the modification and relocation of attachment components such as handrails, maintenance rails and crash barriers are widely utilized to mitigate the unsteady aerodynamic loads acting on the bridge girder. Haque et al. (2016) numerically compared the aerodynamic

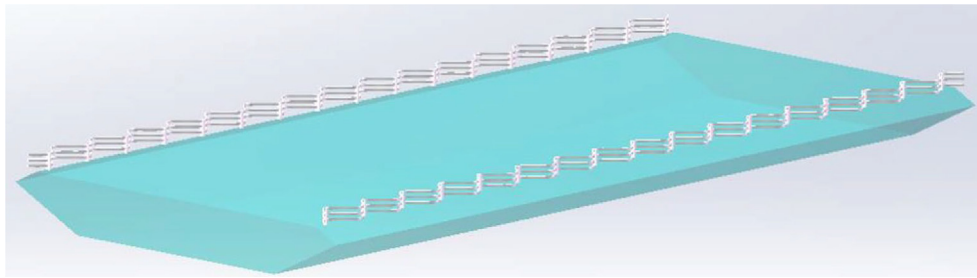


Fig. 29. Description of the sectional bridge model modified with wavy railings (Zhan et al., 2020).

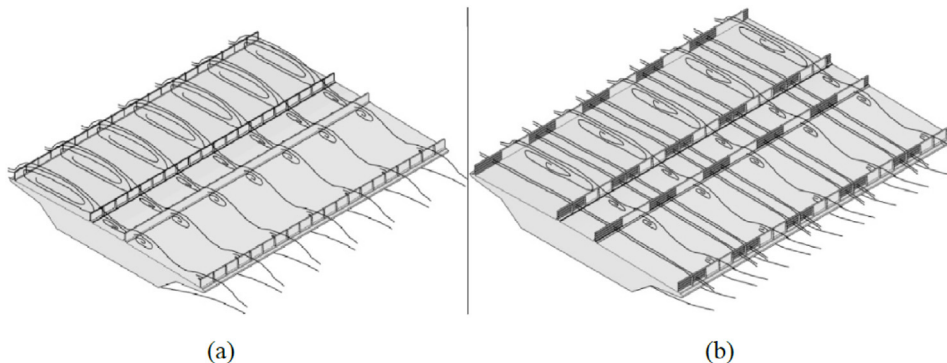


Fig. 30. Flow field associated with two types of sealed traffic barriers: (a) configured continuously along the bridge girder; (b) configured partially along the bridge girder (Bai et al., 2020).

behavior between the solid and perforated handrails on a long-span bridge and concluded that the perforated handrails can reduce the vibration amplitudes, narrow the wake size and improve aerodynamic stability, which is ascribed to the flow moving on the upper surface of the girder, resulting in the disappearance of the upper girder vortex. Qian et al. (2015) pointed out that the handrails with circular cross section have a better capability of VIV suppression than the square cross section. Zhan et al. (2020) proposed a wavy railing consists of vertical posts and wavy crossbars, as shown in Fig. 29, to impose three-dimensional disturbance into the flow field, and trigger the wake second instability. When the wavelength of the wavy railing is 0.5–1 times of girder height, both the vertical and torsional VIV can be eliminated. It should be noted that a proper design of wavy railing is effective to mitigate VIV, however, this configuration is far from practical applications in bridge engineering for it will take the place on the deck. (Bai et al., 2020) put forward an alternative VIV countermeasure, i.e., sealed traffic barrier in Fig. 30, for girders and validated its effectiveness to ameliorate the aerodynamic behavior of bridges via wind tunnel tests. The results showed that the traffic barrier sealed with regular vertical intervals along the girder can effectively eliminate the vertical VIV, and can achieve better performance on VIV suppression than the arc-shaped sealing (Fig. 31). Spoilers installed on the top of handrails with the inclination of 15–45 deg can also mitigate the vertical VIV of a streamlined girder (Hu et al., 2019), as shown in Fig. 32. The spoilers mounted on the barriers provide disturbances into the boundary layer to weaken the correlation between the local aerodynamic force and general vortex-excited force and then repress the vertical VIV. Wang et al. (2011) investigated the influence of various attachment components on the wind-induced vibration of a bridge section model and concluded that the handrail with a high porosity exhibits better aerodynamic stability because it can avoid the separation bubbles which may lead to an increase of VIV amplitude. In terms of the rail of the inspiration car, if it was located far away from the corner of the girder bottom plate, the VIV amplitude will be eliminated to zero approximately. Li et al. (2015) conducted experimental investigations to explore the aerodynamic effect of the maintenance rails on the VIV response of a bridge section model and pointed out that the location of maintenance rails is an important factor for affecting the aeroelastic behaviors. The VIV amplitude experiences a decline when the maintenance rails move to the center of the bottom surface or perpendicularly far away from the bottom surface. Besides, a better improvement of aerodynamic performance can be attained by attaching guide vanes at the maintenance rails.

3.3. Self-issued jet

Although the passive control methods mentioned above are confirmed to be effective for mitigating the VIV response, some unfavorable effects may occur. For instance, the vortex generator can cause a growth of the local pressure and

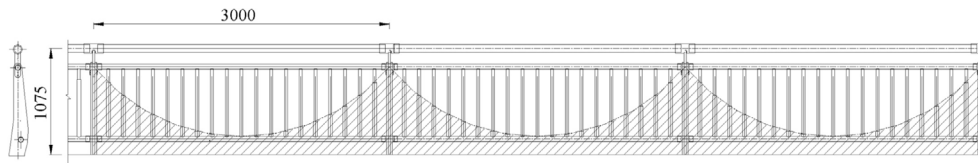


Fig. 31. The schematic diagram of arc-shaped sealing (Bai et al., 2020).

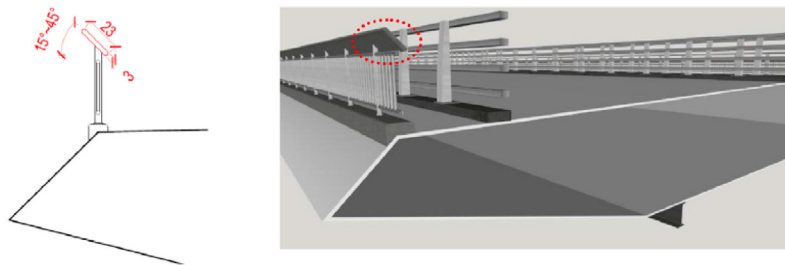


Fig. 32. Description of a section bridge model modified with spoilers of 15–45° inclination (Hu et al., 2019).

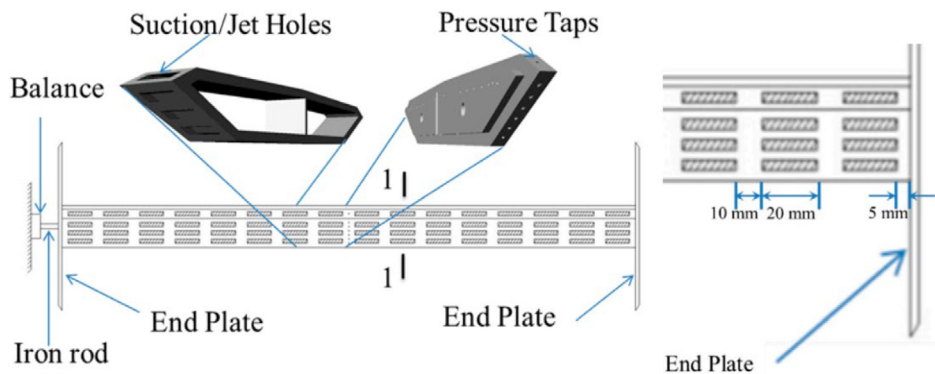


Fig. 33. The segment model of box girder modified with bypass suction and jet holes (Zhang et al., 2017).

surface friction, leading to an augmentation of the drag coefficient of the structure. A jet strategy was introduced into the VIV control of a bluff body. Jets can be deemed as novel vortex generators, namely, air-jet vortex generators, which generate longitudinal vortices to delay the flow separation (Selby et al., 1992) or transfer a two-dimensional flow into a three-dimensional one (Aydin et al., 2010). From the perspective of energy, the air-jet vortex generator strengthens the turbulence and then redistributes the energy of the boundary layer, and no additional drag is generated.

Passive air-jet vortex generators are driven by the pressure difference between the leading and trailing edge of the structure. Based on the ideas provided by Karniadakis and Triantafyllou (1992) and Chen et al. (2015), who introduced passive jet methods to control the VIV of circular cylinder, Zhang et al. (2017) proposed a passive jet method to manipulate the unsteady flow field around the sectional model of the Great Belt East Bridge. The suction and jet holes were set in the windward and leeward inclined panels and were connected through the bypass channels as shown in Fig. 33. The jets are guided into the wake region to suppress the fluctuating aerodynamic force and redistribute the surface pressures on the girder. Furthermore, it also delays the flow separation on the upper surface and reduces the wake width as well as the vortical strength of the shedding vortex.

Chen et al. (2019) further developed this self-suction-and-jet passive control schemes and investigated the influence of the jet location (quantified by the nondimensional suction momentum coefficient J_{suc}) on optimizing aerodynamic performance of the main girder, as shown in Fig. 34. The results demonstrated that this method can suppress the vortex shedding and T.K.E in the wake region, and a large J_{suc} has a better performance on controlling the fluctuating surface pressures and aerodynamic forces. To understand the control mechanism of the passive jets, wind tunnel experiments were proposed to control the VIV (Chen et al., 2019). It was found that the jet location plays a significant role in the control effect. Jets discharged from the slits on the lower half of the lower inclined pane can completely suppress the vertical VIV of the section model, and divide the original lower vortex A (Fig. 5) into two parts: vortex S and vortex A' as shown in Fig. 35. The former located between the bottom corner and slit edge is relatively stable, and the latter vortex is

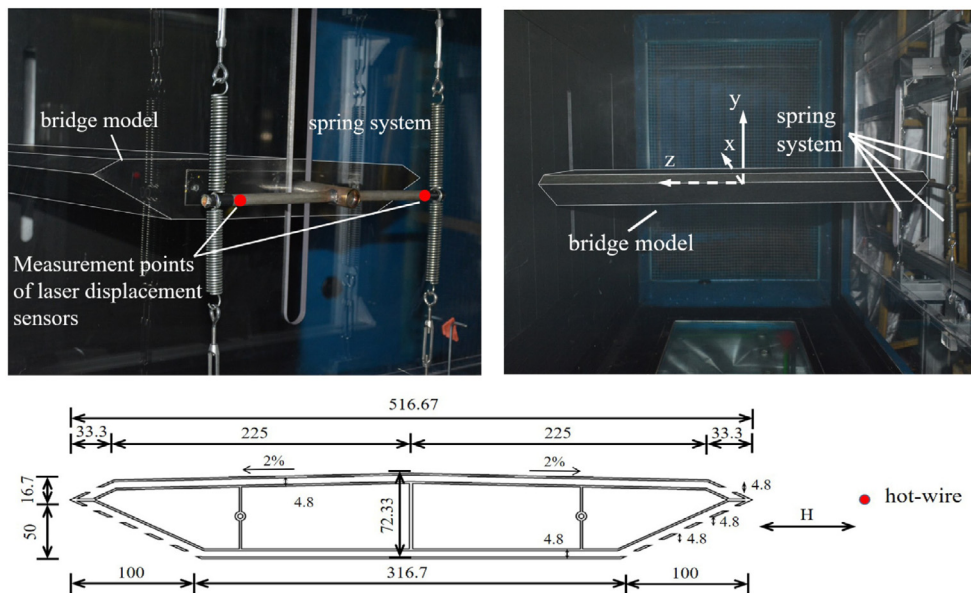


Fig. 34. Spring-supported bridge model in the wind tunnel test and model section with self-suction-and-jet control (Chen et al., 2019).

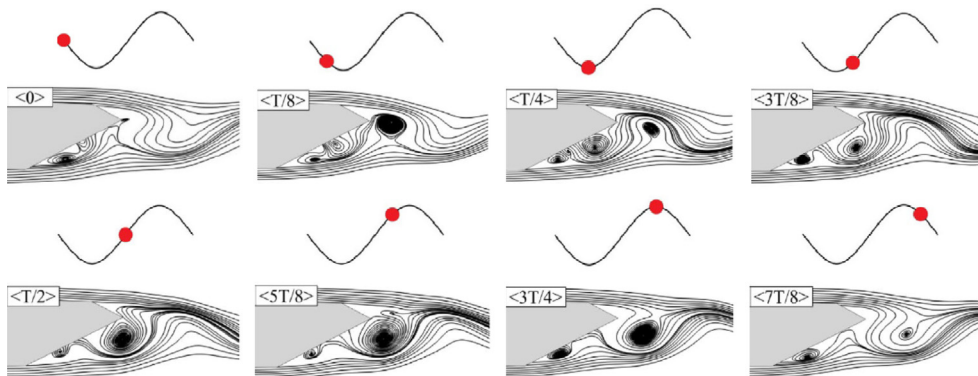


Fig. 35. The vortical evolution behind the trailing edge of the modified box girder at eight different time phases (Chen et al., 2019).

small compared to vortex A and sheds toward downstream with weaker strength. Besides, Mode B streamwise vortices occur in the wake as pictured in Fig. 36, indicating that the jets can trigger three-dimensional instability and contribute to suppressing the VIV (Fig. 36). In the following research, Yang et al. (2020) experimentally studied the AOA effect of the self-issued jet on the VIV of a girder, and pointed out that the self-issued jet performs a better effect when the AOA is positive due to the large pressure difference, and the flutter instability is less affected by this method.

4. Active flow control

Passive control strategy has the advantages of simple structure, low cost, and convenient maintenance, while it is subject to the ambient conditions. For instance, the guide vane is affected by Re effect and AOAs Zhang et al. (2010). To improve the control effectiveness, active control strategies with energy input were proposed to adopt to various external conditions and promote the structural aerodynamic stability (Jahanmiri, 2010; Collis et al., 2004). In this section, active flow control methods for bridge girders are reviewed.

4.1. Attachments

The concept of active attachments was first proposed by Raggett (1987), who installed active winglets on a slender bridge and successfully improved the aerodynamic performance. Boberg et al. (2015) attached active flap arrays to the leading and trailing edges of the girder model to control the wind-induced vibration as shown in Fig. 37, and summarized

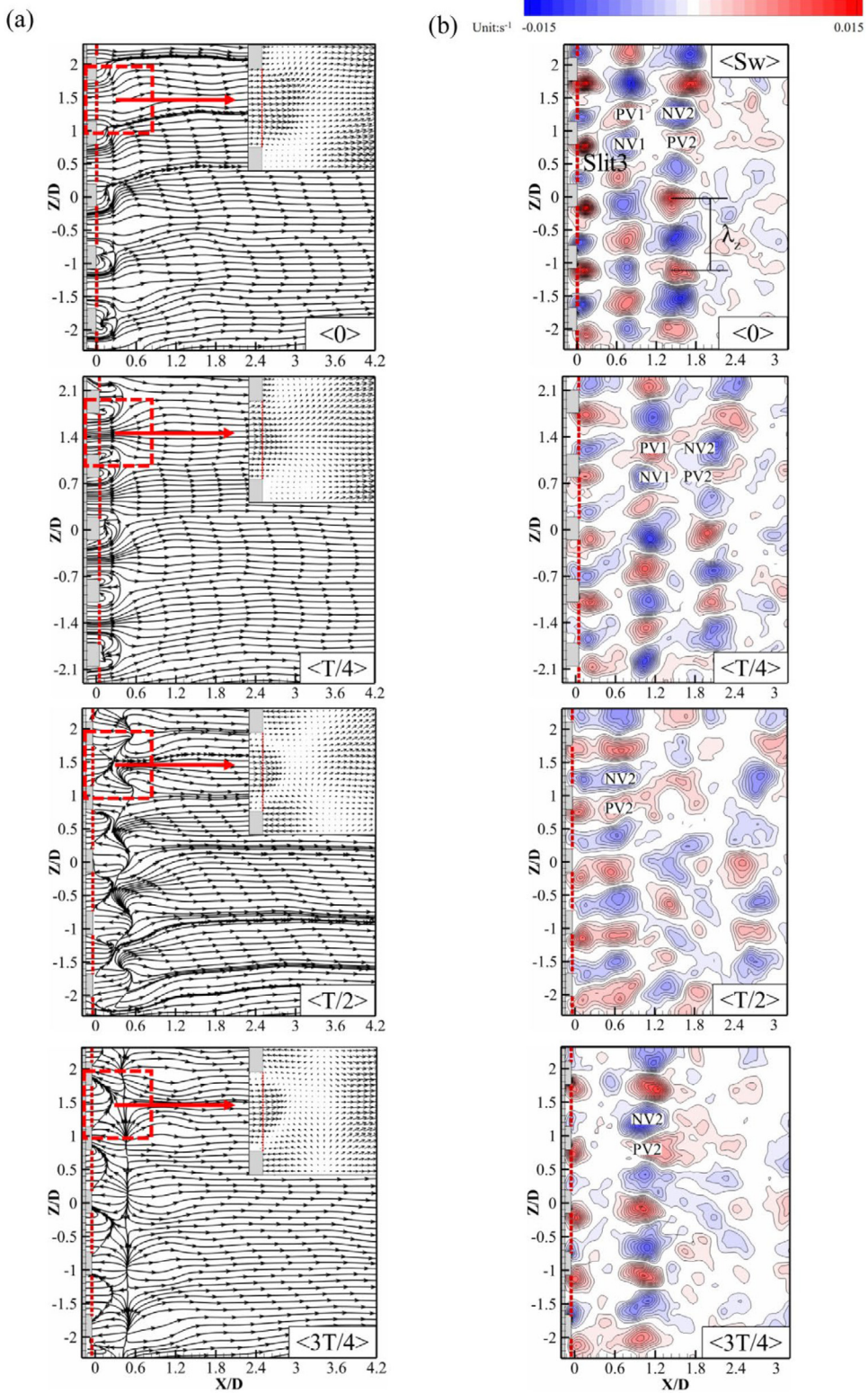


Fig. 36. The flow topology evolution at four different time phases in the modified girder wake, where the Mode B secondary instability was observed (Chen et al., 2019).

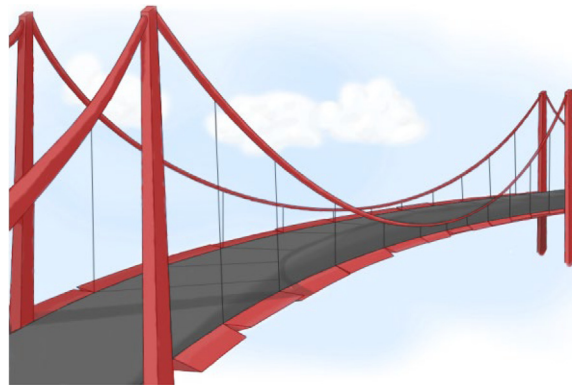


Fig. 37. Concept figure of the bridge girder installed with active flap arrays (Boberg et al., 2015).

that the active arrays have the ability to increase the aerodynamic damping of the sectional model and thus entirely suppress the torsional vibration amplitude.

The active methods above show a satisfactory effect on improving flutter stability. While for VIV, few countermeasures are put forward for bridge girders. Ge et al. (2011) designed an adjustable wind barrier to change the flow structures around the Xihoumen Bridge as illustrated in Fig. 23, and pointed out that the VIV amplitude was remarkably reduced by the wind barrier in a vertical state. However, for a girder in service phase, the vertical windbreak may lead to an increase of the drag coefficient, particularly for the stream-like box girder with an incremental quantity of 185% (Buljac et al., 2017). This adjustable wind barrier also can solve this problem by switching itself from a vertical position to a horizontal one to ensure the bridge safety at a high incoming wind speed. Up to now, the active strategies are still in the research phase and more on-going studies are needed before application.

4.2. Suction and jet

Suction and jet schemes are proved to be effective on VIV suppression. Dong et al. (2008), Naghib-Lahouti et al. (2012) Compared with passive jets aforementioned, the suction/jet generated by flow facilities carries more momentum, resulting in stronger aerodynamic interactions between the jets and the flow field around the structure, as reviewed by Gao et al. (2019a). In this way, the aerodynamic stability of the structure is evidently improved.

Before reviewing the application of active suction and jet control in bridge engineering, it is necessary to introduce the research findings in the field of flow stability theory, which provided a new insight to vortex shedding in a bluff body wake (Chomaz, 2005). Through the flow stability analysis, the sensitive region in the flow field can be easily identified and thus offer a forecast to attain the optimal control effects by imposing small disturbance (Meliga et al., 2014). According to Leu and Ho (2000), the near wake is dominated by absolute instability, which turns into convective instability when it transforms further downstream where the velocity deficits are rapidly filled. It is generally accepted that the global wake instability is dominated by absolute instability in the near wake, and the absolute growth rate is a function of the time-averaged velocity profile of steady unstable flow, so it is significant for VIV mitigation to modify the velocity profiles in the near wake which is the most crucial region to be manipulated. Choi et al. (2008) Both suction and jet control are based on the principle of reducing the absolute instability region by modifying the velocity profile at the near wake.

The active suction/jet strategy has been widely used in the flow control of bluff bodies, while its applications in bridge engineering for controlling the VIV are limited. Zhang et al. (2016) arranged a series of suction holes along the span of the girder to numerically investigate the aerodynamic effect of the active suction on a sectional girder at $Re = 1.31 \times 10^5$, as described in Fig. 38. The suction holes mounted at the lower surface of the girder with a suitable distance can trigger the mode-A three-dimensional instability in the girder wake, resulting in the dislocation and suppression of spanwise vortices. It should be noted that the suction holes close to the trailing corner perform more control effect on suppressing the pressure fluctuations and unsteady aerodynamic forces, indicating this potential for controlling the VIV. In addition, this method significantly modifies the aerodynamic shape by changing the time-averaged velocity distribution in the near-wake region. Tao et al. (2017) numerically and experimentally investigated the control effects of spanwise suction method on the VIV of the girder and found that the VIV control effect of the suction method can contribute to the growth of effective damping. The system stability can be significantly improved by altering the phase difference between the exciting force and lift force of the sectional model. The optimal control effect can be obtained when the suction holes are arranged close to the wake, where the original large-scale vortices are broken into small-scale vortices and the velocity profile is modified observably. Ke et al. (2014) arranged the suction holes on the upper and lower separation points of the upwind girder, as illustrated in Fig. 39. It was found that the upper suction hole can reduce the drag coefficient by nearly 67% at positive AOA, and the lower suction hole can decrease the drag force by approximately 33% at negative AOA. A larger

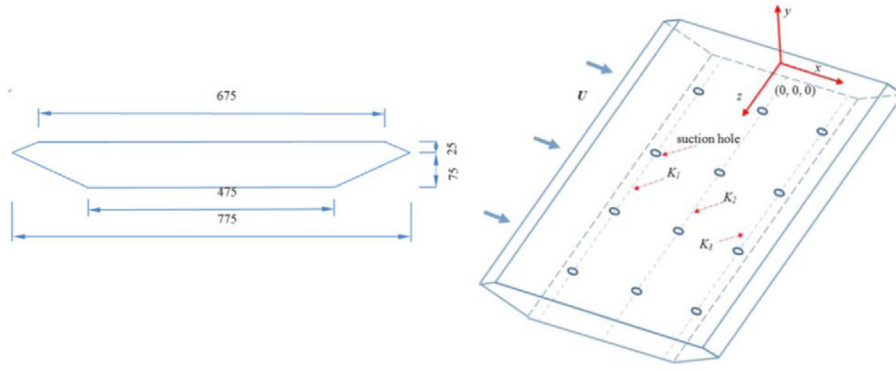


Fig. 38. Configuration of main girder modified with isolated suction holes at the bottom surface (Zhang et al., 2016).

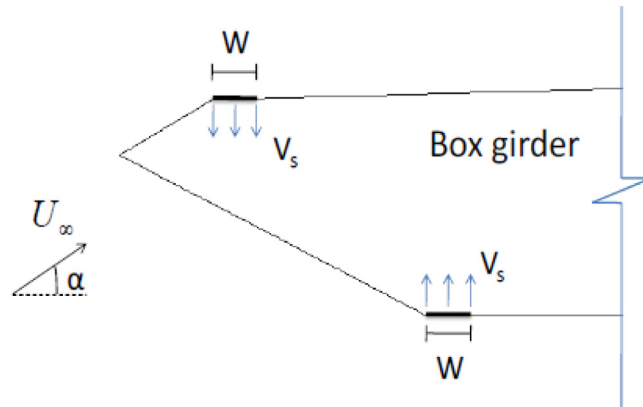


Fig. 39. Configuration of main girder modified with suction holes arranged at the upper and lower separation points (Ke et al., 2014).

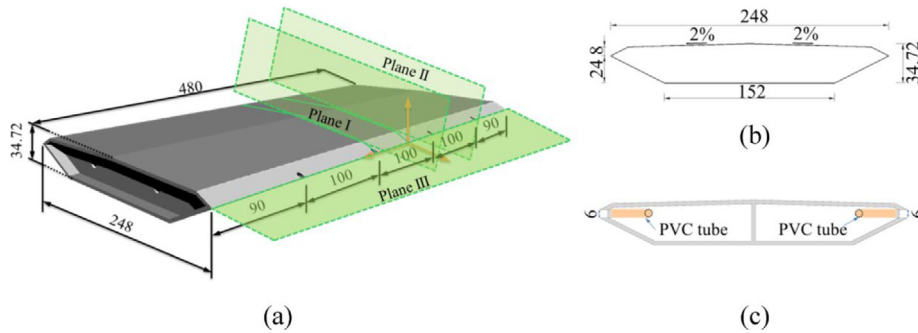


Fig. 40. Configuration of active control method characterized by windward suction and leeward jets (Chen et al., 2020).

suction ratio can effectively suppress the vortex shedding process and reduce the vortex shedding frequency, and thus achieve better control effects. Chen et al. (2020) designed a novel active control approach characterized by leading edge suction and trailing edge jet (LSTJ) on a main girder, as illustrated in Fig. 40. It was observed that the interaction between the active jets and the shear layers can mitigate the T.K.E in the near wake, reduce the strength of the wake vortices and push the unsteady flow structure further downstream. In this way, the stability characteristics were evidently modified by narrowing the area of absolute instability, and the unsteady aerodynamic force acting on the girder were effectively suppressed.

5. Conclusion and perspectives

A selective review of vortex-induced vibration (VIV) and its aerodynamic control strategies for the long-span bridge are discussed from the perspective of vortex dynamics. Firstly, the VIV mechanism is discussed for different types of girders (single-box and twin-box) based on field monitoring, wind tunnel test, and computational fluid dynamics (CFD) simulation results. It is concluded that the VIV of main girders can be excited by the motion-induced vortex, von Kármán vortex shedding, and gap vortex, depending on the surrounding flow patterns and the dynamic vortex evolutions. In terms of active and passive ones, the aerodynamic control strategies for vortex manipulation and VIV mitigation are then summarized and discussed in detail. The effects and underlying mechanism of different types of control methods are also discussed as a focus of this review.

It should be noted that, with the increasing demand of constructing very long suspension bridges (with a main span of 3000 m or even longer), the separated-body type of main girders is attracting the focus of bridge engineers. Besides the planned Messina strait bridge (Italy, with a main span of 3300 m), the new Xihoumen rail-cum-road bridge, which is under construction, also adopt a separated three-box as the cross section of the main girder. It is generally accepted that a separated-body is helpful to enhance the flutter instability of long-span bridges. However, due to the existence of gaps and complex gap flows, a cross-section with separated-body leads to much more complex flow–structure interactions. Meanwhile, because of the shedding of gap vortices, a separated-body girder is more sensitive to VIV, as in the case of old Xihoumen suspension bridge. Therefore, understanding the vortex dynamics and mechanism of VIV of a three-separated-box-girder and developing useful aerodynamic control strategies are still in urgent demand to the community of bridge wind engineering.

Though the vortex dynamics-based studies of VIV are still growing and ascendant, further efforts are also called to the modeling of the fluid–structure interaction of VIV, and the prediction and recognition of VIV events of the prototype bridge (see Li et al. (2017) for instance). In addition, mitigation of VIV by means of aerodynamic control cannot be overlooked in the sense that the VIV of bridge girders is caused by the surrounding flow. With proper configuration, some classical control strategies can be reinvigorated for bridge engineering applications. For instance, the barriers and attachments can be designed elaborately to be vortex generators to suppress the vortex shedding and mitigate the VIV amplitudes of long-span bridges, as in the case of Humen suspension bridge. On the other hand, some new flow control methods can be introduced to VIV mitigation of bridge girders, such as active/passive jets and ionic wind at proper positions. These schemes are applicable because the main girder of long span bridges is usually with a very large geometric scale and provides sufficient room to realize the control process. To sum up, flow control methods are promising in VIV mitigation of long-span bridges but it is still a challenge for practical engineering application. These possibilities remain to be explored and more research efforts need to be devoted.

CRedit authorship contribution statement

Donglai Gao: Methodology and organization, Funding acquisition, Data and figures, Writing and editing of the manuscript. **Zhi Deng:** Data and figures, Text preparation. **Wenhan Yang:** Data and figures, Text preparation. **Wenli Chen:** Supervision, Methodology, Funding acquisition, Editing of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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